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THEORETICAL/REVIEW

Bits of confidence: Metacognition as uncertainty reduction

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Abstract

How do people know when they are right? Confidence judgments – the ability to assess the correctness of one's own decisions – are a key aspect of human metacognition. This self-evaluative act plays a central role in learning, memory, consciousness, and group decision-making. In this paper, I reframe metacognition as a structured exchange of information between stimulus, decision-maker (the actor), and confidence judge (the rater), akin to a multi-agent communication system. Within this framework, the actor aims to resolve stimulus uncertainty, while the rater seeks to infer the accuracy of the actor's response. Applying techniques from information theory, I develop three novel measures of metacognitive efficiency: *meta-*$\mathcal{U}$, *meta-*$\mathcal{KL}$, and *meta-*$\mathcal{J}$. These indices are derived from entropy and divergence principles, and quantify how effectively confidence judgments transmit information about both external stimuli and internal decisions. Simulations show that these measures possess several advantages over traditional signal detection theory metrics such as *meta-d′* and the *M*-ratio, including more interpretable scaling, robustness to performance imbalances, and sensitivity to structural constraints. By formalizing metacognitive sensitivity as an information-processing problem, this framework offers a unified, theoretically grounded approach to studying confidence and sheds light on the sources of metacognitive inefficiency across individuals and contexts.

Keywords Metacognition · Confidence · Information theory · meta-d' · M-ratio · meta-I

Introduction

A primary topic in psychology and neuroscience concerns the ability of humans to assess the correctness of their own decisions via confidence ratings (Peirce & Jastrow, 1884; Henmon, 1911; Lau, 2022; Metcalfe & Shimamura, 1994; Fleming & Dolan, 2012; Fleming, Weil, Nagy, Dolan, & Rees, 2010). This self-reflective act is essential for survival (see Fleming, 2024), but it is also fraught with confusion and uncertainty (Pouget, Drugowitsch, & Kepecs, 2016). Metacognition plays a central role in learning (Meyniel, Schlunegger, & Dehaene, 2015), memory (Koriat & Goldsmith, 1996; Koriat, 1997), consciousness (Koriat, 2000), action (Persaud, McLeod, & Cowey, 2007; Van den Berg, Zylberberg, Kiani, Shadlen, & Wolpert, 2016), psychiatric abnormalities (Hoven et al., 2019), and cooperative group decision (Bahrami et al., 2010). Self-assessment and confidence are also crucial in searching for new information (Pleskac & Busemeyer, 2010; Schulz, Fleming, & Dayan, 2023; Schulz, Rollwage, Dolan, & Fleming, 2020). For example, when a student is preparing for an exam, she must continually monitor her own level of knowledge. If she believes that she has acquired a sufficient amount of information, she can stop her learning; otherwise, she will continue her preparations.

In the laboratory, various tasks have been developed to measure metacognition, including self-reports (Craig, Hale, Grainger, & Stewart, 2020), opt-out responses (Kepecs & Mainen, 2012), post-decision wagers (Persaud et al., 2007), and confidence rating (Fleming & Lau, 2014; Fitousi, 2024). An important determinant of self-assessment is dubbed *metacognitive sensitivity* (Fleming & Lau, 2014). This concept refers to the participant's ability to discriminate between correct and incorrect responses based on confidence ratings. Procedures that test for confidence are often dubbed *second-order* or *type 2* tasks (Clarke, Birdsall, & Tanner Jr, 1959; Galvin, Podd, Drga, & Whitmore, 2003). This in contrast to *first-order* or *type 1* tasks, whereby participants are asked to perceive and judge the stimuli on their own right. One popular measure of type 2 sensitivity records how often confidence ratings are congruent with accuracy (Persaud et al., 2007). Another gauges the *phi* (Kornell, Son, & Terrace, 2007),

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or gamma (Nelson, 1984) correlation between accuracy and confidence. More sophisticated approaches – such as $meta\text{-}d'$ (Maniscalco & Lau, 2012, 2014) – are based on signal detection theory (SDT, Green & Swets, 1966; Macmillan & Creelman, 2004), incorporating distributional assumptions (Kunimoto, Miller, & Pashler, 2001; Galvin et al., 2003; Clarke et al., 1959). The main goal of these measures is to quantify the degree to which the observer is efficient in using the information in the signal to predict their decisions. Notably, a viable measure of metacognitive efficiency should accommodate an inherent tension existing between type 1 and type 2 sensitivity. Trivially, type 2 sensitivity can increase as the accuracy of type 1 performance improves. To solve this problem, measures of *metacognitive efficiency*, such as $meta\text{-}d' - d'_0$ or the M-ratio (Maniscalco & Lau, 2012, 2014), have been constructed. These measures are of immense theoretical importance and among the most popular vehicles for assessing metacognitive abilities. A wealth of studies, building on these measures, have shown that confidence ratings in many tasks reflect sub-optimal metacognitive efficiency (Shekhar & Rahnev, 2021b). However, these measures are not without their limitations (Dayan, 2023). The SDT-based measures in particular: (a) require distributional assumptions, (b) are applicable only to ratio-scale variables, (c) do not scale with the number of stimuli, responses, or confidence rating alternatives, and (d) are applicable only to experimental designs with two stimuli and two responses.

The present effort circumvents these limitations by offering a novel approach to metacognitive efficiency and confidence that relies on principles of information theory (Shannon, 1948; Shannon & Weaver, 1949). Metacognition is viewed here as a complex system of communication (see Fig. 1) involving three roles: the experimenter/stimuli, the actor, and the rater (Garner, 1960, 1962, 1974; Eriksen & Hake, 1955). The goal of both actor and rater is to maintain reliable communication over an unreliable channel (Shannon, 1948; MacKay, 2003), by reducing the level of uncertainty in their communication. The role of the experimenter in this setting is often unacknowledged, but is of immense impact on the observed results (Algom, Dekel, & Pansky, 1996; Algom & Fitousi, 2016; Fitousi & Algom, 2006; Garner, 1962, 1974; Melara & Algom, 2003; Melara & Mounts, 1993). By deciding on the number of stimuli, responses, and confidence rating buckets, the experimenter determines the degree of overall uncertainty in the system, and therefore sets lower and upper bounds on the amount of information that can be transmitted from the stimuli to the actor, and from the actor to the rater (Garner, 1962, 1974; Killeen & Taylor, 2001). The system is characterized by a total amount of uncertainty (Garner, 1962; Fitousi, 2013; Eriksen & Hake, 1955), that can be partitioned into various meaningful components (Garner & Morton, 1969; Shekhar & Rahnev, 2021b).

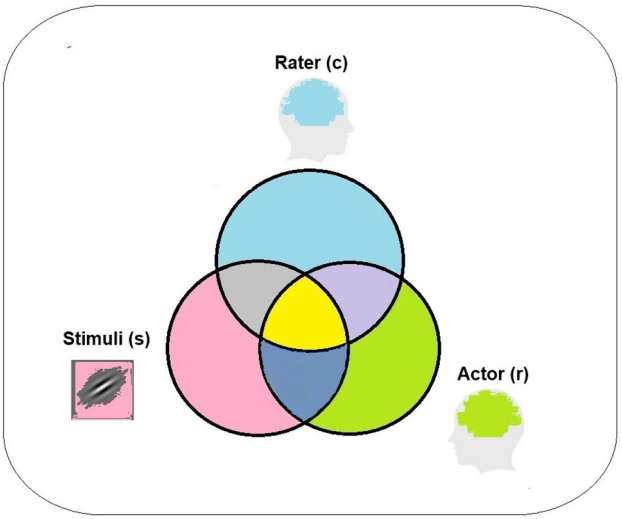


Fig. 1 Illustration of the stimuli–response–confidence communication system. Uncertainty in stimuli (s) is represented by a *pink circle*. Uncertainty in actor's response (r) is represented by a *green circle*. Uncertainty in rater's confidence (c) is represented by a *blue circle*. An overlap between circles represents their mutual information. The novel mutual information measure of metacognitive efficiency gauges the mutual information between confidence and the stimulus-response variables. The figure is a modified version of a figure from Fitousi (2025b), published under the terms of the Creative Commons Attribution License (CC BY 4.0). © 2025 by the authors; published by MDPI, Basel, Switzerland

This type of analysis can shed new light on the relations ensuing between cognition and metacognition, as well as on the underlying exchange of information between stimuli/experimenter, actor and rater. The present work introduces several converging measures of cognitive and metacognitive efficiency: $meta\text{-}\mathcal{KL}$, $meta\text{-}\mathcal{J}$, and $meta\text{-}\mathcal{U}$. These are based on the *Kullback-Leibler* divergence (Kullback, 1959), the *Jeffreys* divergence (Jeffreys, 1992; Pearl, 1990), and the mutual information of three variables (Garner, 1962; Cover & Thomas, 1991; Pearl, 1990). These measures are non-metric, and as such have several desirable characteristics: (a) they do not require distributional assumptions, (b) are applicable to nominal scale variables, (c) converge with other information and SDT measures, (d) scale with the number of stimuli, responses, and rating alternatives, and (e) can be applied to experimental designs with more than two stimuli and responses. A key insight from the present investigation is that metacognitive efficiency depends on both the structural information constraints (i.e., total uncertainty) inherent to the experimental design in question and on the individual differences in the actors' and raters' channel capacity.

The advancement of an information theory framework of metacognition and confidence can be fruitful and timely. In recent years, there has been a resurgence of interest in information-theoretic approaches to perception and cognition (Algom & Fitousi, 2025; Feldman, 2021; Fitousi,

2013, 2023; Fitousi & Algom, 2024; Friston, 2010; Hirsh, Mar, & Peterson, 2012; Killeen & Taylor, 2001; Knill & Pouget, 2004; Norwich, 1993). This is not surprising given that behavior and brain are characterized by uncertainty (Pouget et al., 2016), and information theory (Shannon, 1948; Shannon & Weaver, 1949) offers powerful tools for dealing with such entropy (Friston, 2009). In the area of confidence and metacognition, a recent paper by Dayan (2023) has ignited much interest in this type of analysis. Dayan (2023) succinctly summarized the suitability and utility of an information-theoretic approach to the study of confidence: "the problem of confidence is inherently one of information – that the actor has about the true state of the stimulus; and that the rater has about the same quantity and about what the actor used. It therefore seems appropriate to use the methods of information theory to judge the relationship between the stimulus, the actor, and the rater" (p.11).

Developing information-theoretic conceptions and measures

There are several theoretical and practical justifications for developing information-theoretic approaches to metacognition. First, the concept of information itself shows up frequently in theoretical discussions of metacognition, either as an explanandum or an explanan. For example, it is commonly assumed that metacognition guides the organism on whether to seek further information (Schulz et al., 2020, 2023). Moreover, researchers often argue that the amount of information the rater can extract from the actor's response is bounded from above by the amount of information extracted by the actor from the stimuli (Maniscalco & Lau, 2012). But rarely do researchers define these information quantities in rigorous information theory terms. This is unfortunate, because information theory can provide exact formulas for capacity limitations, distortion rate, as well as estimations of upper and lower bounds on performance (Cover & Thomas, 1991; MacKay, 2003). These are not readily available in other, more popular, theoretical frameworks (e.g., SDT). Information theory measures can be derived without making any distributional assumptions. This is particularly valuable since many model-based measures of type 1 and type 2 performance, such as d', meta-d', or M-ratio (Maniscalco & Lau, 2012) rely heavily on parametric assumptions. The problem is more severe in estimations of type 2 sensitivity, for which distributional assumptions have been shown to be highly dubious (Galvin et al., 2003). Unlike existing measures, information-theory measures are bias-free, can be applied to nominal scale variables, and do not make any assumptions regarding order or relative magnitude of ratings scales, stimuli, or responses alternatives.

The parametric assumptions underlying existing model-based measures pose additional problems. There is no sound rationale for expecting empirical confidence to fit exactly to a type 1 decision process (Dayan, 2023). Moreover, these measures are susceptible to decisional bias (Maniscalco & Lau, 2012, 2014), which poses a serious statistical challenge in estimating their fit to data. Furthermore, existing measures of metacognitive efficiency violate the necessary condition of independence from metacognitive bias (Xue, Shekhar, & Rahnev, 2021). Information theory measures hold the promise of avoiding these limitations. Another crucial point is that both model-based and model-free measures of metacognitive efficiency (Shekhar & Rahnev, 2021a) do not accommodate changes in the granularity of type 2 confidence ratings, nor do they scale with the number of type 1 stimuli or responses. According to principles of information theory (Attneave, 1959; Garner, 1962), the total amount of uncertainty in a system increases with the number of stimuli, responses, and confidence rating alternatives. Thus, from an information-theoretic perspective, a rater who is capable of making a fine discrimination with an eight-point rating scale should be considered more sensitive than one who operates with a four-point rating scale. This in contrast to existing SDT models where the exact confidence ratings are seen as something that is a product of where the confidence criteria happened to be placed, with no placement being seen as necessarily superior to another (unless there's something that the subject is trying to optimize; see Maniscalco, Charles, & Peters, 2024). The present approach views this as a limitation of existing models. The scale resolution should not be modeled separately from metacognitive sensitivity. The same argument applies to the granularity of response and/or stimuli alternatives in type 1 tasks. Neither meta-d', meta-d' - d_0, nor M-ratio takes these important characteristics into consideration.

It should be noted that the issue of stimuli/response/confidence rating granularity is only one facet of the more general problem of uncertainty. It has been shown that the degree of uncertainty in the experiment, even in dimensions that are irrelevant to the current decisions, can modulate confidence judgments (De Gardelle & Mamassian, 2015; Fetsch, Kiani, Newsome, & Shadlen, 2014; Spence, Dux, & Arnold, 2016; Spence, Mattingley, & Dux, 2018). Notably, Rahnev and Fleming (2019) have found that the staircase experimental procedure led to an inflation in both model-based and model-free measures of metacognitive sensitivity, such that the amount of stimulus variability experienced by the participant correlated with the size of that inflation. Many observations in the confidence literature (Navajas et al., 2017; Spence et al., 2018), including this one, can be better understood within an information theory framework, one that takes into consideration the total and individual components

of uncertainty in the complex stimuli-response-confidence communication system.

The impact of stimulus and response uncertainty on discriminability has been studied extensively by pioneers of information theory in psychology, such as Garner (1962) and Miller (1956), especially in the context of the absolute identification task (Garner & Hake, 1951; Miller, 1956). A key insight from decades of work within this information-theoretic framework (Attneave, 1959; Garner, 1962; Garner & Hake, 1951; Algom & Fitousi, 2016; Melara & Algom, 2003; Fitousi, 2013; Killeen & Taylor, 2001) is that the outcome of any psychological experiment is ultimately shaped by the underlying informational structure (Garner, 1962, 1974). An information-theoretic approach to metacognition can explain phenomena that other approaches, such as SDT (Macmillan & Creelman, 2004), cannot. This includes the findings that increasing the number of response alternatives in first-order (Garner, 1960) or second-order (Dayan, 2023) tasks can effectively or potentially improve discriminability.

Furthermore, most SDT approaches are restricted to two-choice type 1 tasks, and cannot be readily applied to tasks with multi-alternative choices (Li & Ma, 2020). There are experimental designs that may require more complex structures, with more than two stimuli or/and more than two responses in the primary task. For example, testing the metacognitive efficiency of facial emotion recognition (Kelly & Metcalfe, 2011), requires an experiment with the basic six facial emotions (Ekman, 1976). On each trial, the actor identifies the emotion on a presented face, and then the rater is giving her confidence on an eight-point scale point with respect to the actor's response. With all their ingenuity, SDT canonical measures cannot be deployed in such a complex experimental design. Recently, Shekhar and Rahnev (2021b) have noted that: "making...complex models identifiable, requires experiments with more complex designs – such as having tasks with multiple alternatives..." (p. 20). The upshot is that more general techniques and measures of metacognitive sensitivity should be advanced. This goal is accomplished in the present study.

Finally, existing measures of metacognitive sensitivity are based on the assumption that the rater's confidence reflects the probability of the actor's response being correct (Maniscalco & Lau, 2012). However, several studies (Adler & Ma, 2018a, 2018b; Denison, Adler, Carrasco, & Ma, 2018; Li & Ma, 2020; Navajas et al., 2017) using two-choice tasks showed that most raters do not compute the posterior probability of the actor's being correct, but only approximate this quantity using a heuristic. Moreover, several of these studies provide evidence for the important role of uncertainty in confidence computations. For example, Navajas et al. (2017) and Denison et al. (2018) found that confidence reflected the uncertainty in the stimuli. In yet another study, with three-choice tasks, Li and Ma (2020) found that the raters' confidence was best accounted for by the difference in the posterior probabilities of the two most likely alternatives. Information theory measures can circumvent these difficulties because they take into consideration the various sources of uncertainty in the experiment, assuming a nominal scale of variables. A second reason for advancing information-theory measures is that for some stimuli, there is no straightforward way of determining the correctness of the actor's response (Thurstone, 1927). For example, actors can meaningfully judge the beauty of an artwork (Schwarz, 2018), or the legibility of a pattern of handwriting (Garner, 1960). Still, the rater can assess the actor's consistency and discriminability for these tasks. Thus, one should seek to develop measures that can gauge metacognitive efficiency without recourse to what a "correct" or an "incorrect" response might mean. Information theory measures meet these requirements because they apply to nominal scale variables and are therefore agnostic to the semantic content of transmitted information.

Roadmap to the present paper

The remainder of the paper is structured as follows. I begin with a brief introduction to two prominent computational models of metacognition and confidence: the $meta\text{-}d'$ (Maniscalco & Lau, 2012), and the second-order model (Dayan, 2023; Fleming & Daw, 2017; Jang, Wallsten, & Huber, 2012). These models are elaborated upon because they serve later as benchmarks for validating the new information measures of metacognitive efficiency. I then provide a concise introduction to information theory. Readers who are well-versed in information theory can skip this section.

I then introduce the $meta\text{-}\mathcal{I}$ proposed by Dayan (2023), and start developing my own novel information theoretic measures of metacognitive sensitivity. The first two measures: $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$, are based on the Kullback–Leibler (Kullback, 1959), and Jeffreys (1992) divergences. These measures are initially validated against the standard SDT measures, including $meta\text{-}d'$ and M-ratio, using simulations with the $meta\text{-}d'$ model (Maniscalco & Lau, 2012).

A third measure, $meta\text{-}\mathcal{U}$, is then developed, based on the mutual information of three variables (Pearl, 1990; Cover & Thomas, 1991). The measure is derived in two independent ways. The first approach deploys a technique known as *multivariate uncertainty analysis* (Garner, 1962), which decomposes the total uncertainty in a three-dimensional stimuli-response-confidence matrix. The second approach is based on a canonical analysis (MacKay, 2003). The derivations from the two approaches converge on the same quantitative measure, but offer a complementary view on how metacognitive efficiency can be defined and computed. Importantly, the $meta\text{-}\mathcal{U}$ measure captures the same quantity

as *meta-*$\mathcal{I}$, but is computed differently. Moreover, it is of a broader generality because it can be applied to experimental designs with more than two stimuli and two responses. In addition, the computations of *meta-*$\mathcal{U}$ do not depend on response accuracy.

Subsequently, *meta-*$\mathcal{KL}$, *meta-*$\mathcal{J}$, and *meta-*$\mathcal{U}$ are validated through a series of simulations with the second-order model (Fleming & Daw, 2017). First, I demonstrate the lawful relations between these measures and SDT-based measures. Then, I capitalize on a powerful prediction from the second-order model, according to which increasing the correlation between actor and rater should reduce metacognitive efficiency (Fleming & Daw, 2017). I show that all three information-theory measures adhere to this prediction. Furthermore, I offer a quantitative explanation for this phenomenon from the perspective of information theory and derive a novel set of predictions. These are corroborated in a dedicated series of simulations. Finally, I demonstrate how information-theory measures scale with the granularity of confidence scales.

The *meta-d′* model

The *meta-d'* model (Maniscalco & Lau, 2012) assumes that type 2 metacognitive efficiency is constrained by type 1 sensitivity. In particular, the rater's ability to correctly reflect on the actor's decision is limited from above by the information available in the actor's type 1 task. Thus, a hypothetical type 1 d' index – dubbed *meta-d'* – can be expressed in the same units as the actor's actual d' performance. The key elements of the model are the discrete probability distributions $P(\text{confidence} = y \mid \text{Correct})$ and $Q(\text{confidence} = y \mid \text{Incorrect})$, which describe confidence ratings given correct and incorrect responses, respectively (see Fig. 2). These are derived from the probability of rating confidence y by dividing the corresponding frequency by the number of correct (incorrect) trials. A metacognitively sensitive rater can efficiently distinguish between correct and incorrect trials by giving high confidence ratings for correct responses and low confidence ratings for incorrect responses. The degree of overlap between the two distributions reflects the rater's type 2 sensitivity, and can be used to compute an ideal d' (the so-called *meta-*d') through type 2 ROC analysis.

A viable measure of metacognitive efficiency should take into consideration both type 1 and type 2 performance (Rahnev & Fleming, 2019). In particular, the rater's sensitivity in type 2 tasks should be related to the actor's sensitivity in type 1 tasks. A rater who is performing successfully with respect to an uncertain actor is far more efficient than a rater who is performing successfully with respect to a certain actor. Thus, we cannot compare the two raters. The latter is far more efficient than the former. The comparison of the observed type 1 sensitivity d'_0 with the hypothetical *meta-d'* provides a relative measure of metacognitive efficiency by subtraction (*meta-d'*-d'_0) or division (*meta-d'*$/d'_0$). The rater's efficiency is optimal when *meta-d'* $= d'_0$ (M-ratio = 1). In this case, the rater capitalizes on all the available information from the signal; the rater is hypo-sensitive when *meta-d'* $< d'_0$ (M-ratio < 1). The rater is hyper-sensitive when *meta-d'* $> d'_0$ (M-ratio > 1). Notably, the rater in the *meta-*d' model can be considered as "naïve Bayesian," because his type 2 confidence ratings are based on the same information as the type 1 action. In this sense, the *meta-d'* model is a first-order model (Fleming & Daw, 2017).

The second-order model of metacognition

The *second-order model* of metacognition has been widely applied to model metacognitive processes (Jang et al., 2012; Fleming & Daw, 2017; Schulz et al., 2023). Figure 3 illustrates one realization of the model. The model assumes an actor (*act*) who performs in a two-choice discrimination task, and a confidence-rater (*conf*), who evaluates the correctness of the actor's response. The separation between the two is consistent with studies showing double dissociation between accuracy and confidence (for review see, Shekhar & Rahnev, 2021b). Both actor and rater are given internal samples X_{act} and X_{conf} of world state d (e.g., a face is either happy or angry). These are modeled as a bivariate Gaussian with a mean vector:

$$\mu = \begin{bmatrix} \mu_{Xact} \\ \mu_{Xconf} \end{bmatrix} \tag{1}$$

and a variance-covariance matrix Σ:

$$\Sigma = \begin{bmatrix} \sigma^2_{act} & \rho\sigma_{act}\sigma_{conf} \\ \rho\sigma_{act}\sigma_{conf} & \sigma^2_{conf} \end{bmatrix} \tag{2}$$

where the subscripts act and $conf$ index the parameters for the marginal distributions on evidence for actor's and rater's sampling distributions. The three parameters in the covariance matrix are: σ_{act}, σ_{conf}, and $\rho\sigma_{act}\sigma_{conf}$. The first two control the noise in the signals for actor and rater, respectively. The ratio of noises in the actor and rater signals influences metacognitive sensitivity. As the ratio $\sigma_{act}/\sigma_{conf}$ increases, metacognitive sensitivity gets better. The third parameter, $\rho\sigma_{act}\sigma_{conf}$, governs the correlation between the two samples. When $\mid \rho \mid > 0$, the rater's and actor's samples are associated to some degree. In the extreme case, $\mid \rho \mid = 1$, the two are perfectly correlated. When $\rho = 0$, the two are independent. From an information-theoretic perspective, exchange of information between actor and rater have quantifiable and explainable implications on the rater's

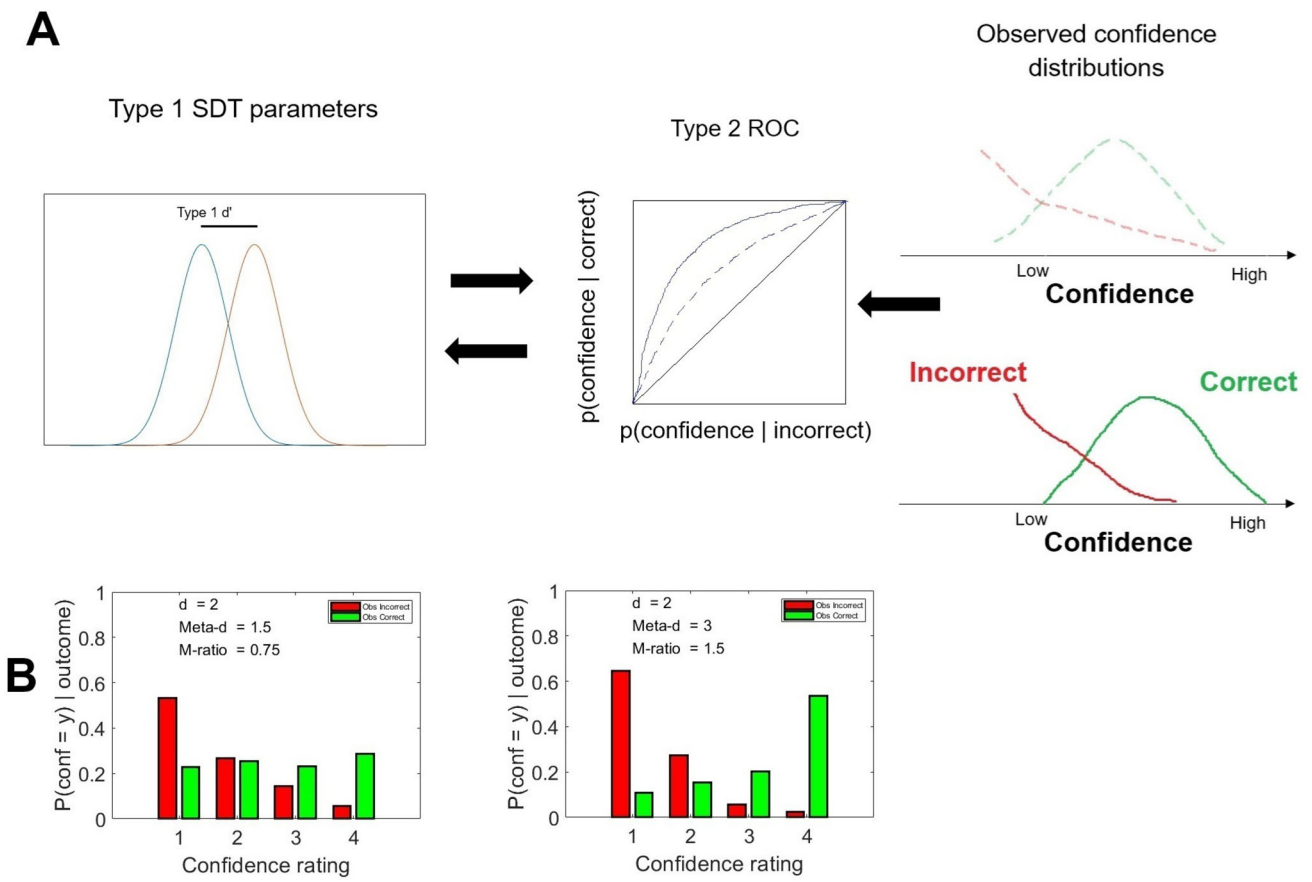


Fig. 2 The *meta-d′* model. **(A)** Right panel presents theoretical confidence-rating distributions conditional on correct and incorrect responses. A rater with good metacognitive sensitivity gives higher confidence ratings when the actor is correct, and lower confidence ratings when the actor is incorrect. Large overlap between those distributions reflects low metacognitive sensitivity (distributions with *dashed lines*), while the converse also holds true (distributions with *full lines*). A type 2 ROC curve (*middle panel*) is used to measure the overlap between correct and incorrect distributions. The theoretical type 2 ROC curve is generated by an equal-variance SDT model. Second-order hit rates and false alarms are expressed in the divergence units of the type 1 model d'. The model looks for the theoretical type 1 d' that best fits the observed confidence data (Maniscalco & Lau, 2014). This parameter is called *meta-d′*, and it is compared to the observed type 1 d'. When $meta\text{-}d' < d'_0$ the observer is hyposensitive, when $meta\text{-}d' > d'_0$ the observer is hypersensitive. **(B)** Examples for confidence-rating distributions simulated with $d' = 2$ and $meta\text{-}d' = 1.5$ (*left panel*), or $meta\text{-}d' = 3$ (*right panel*)

metacognitive sensitivity. One goal of the present effort is to demonstrate how this can be done.

The rater's task is to indicate how confident she is in the correctness of the actor's response. This amounts to the posterior probability that the actor's action a was correct given the inferred state of the world d. Computationally, this amounts to the rater inferring the state of the decision variable X_{act} from the confidence variable X_{conf}:

$$z = P(a = d|X_{conf}, a, \Sigma) = \begin{cases} P(d = 1|X_{conf}, a, \Sigma) & \text{if } a = 1 \\ 1 - P(d = 1|X_{conf}, a, \Sigma) & \text{if } a = -1 \end{cases} \quad (3)$$

First-order and *post-decisional* models are nested within this second-order model. In the first-order model, the rater makes an inference based on the same evidence sampled by the actor. In this case, the model reduces to the parameters of X_{act}. In the post-decision model, the rater samples again from the actor's distribution and computes the posterior probability that the actor's action was correct.

Basics of information theory

This section provides a brief introduction to basic formulas and notations in information theory. A comprehensive introduction can be found in classic textbooks (Cover & Thomas, 1991; MacKay, 2003). Assume a discrete variable x. The *Shannon information* $H(x) = -\sum p(x) \log p(x)$ measures the average *entropy* (or uncertainty) associated with this variable. It is maximal when all categories of x

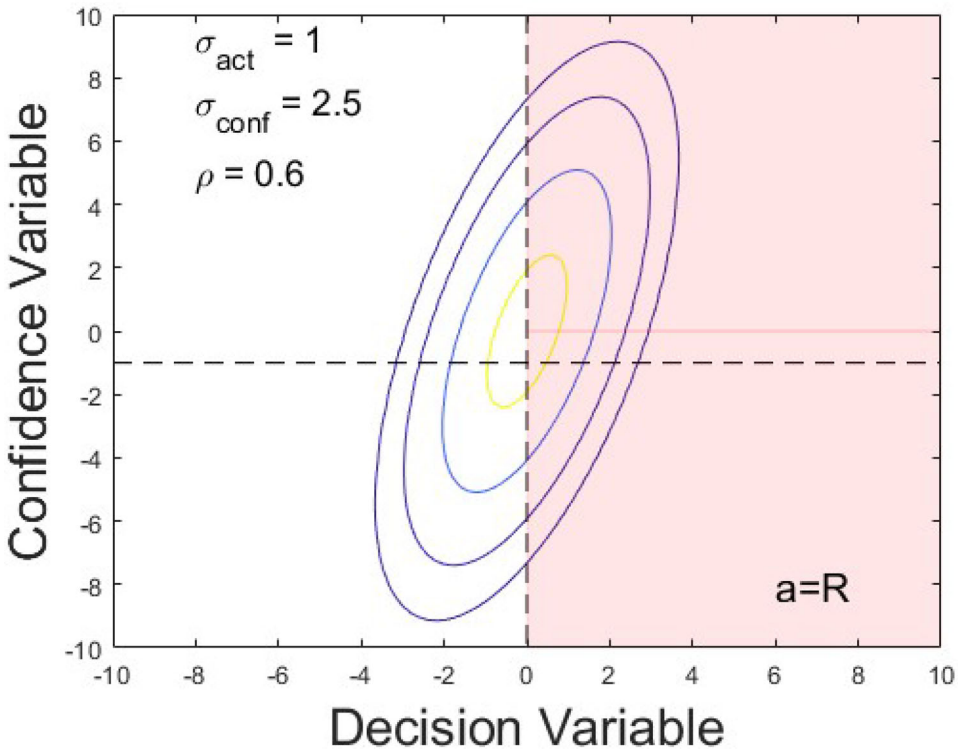


Fig. 3 The second-order model. A bivariate Gaussian distribution represents the joint likelihood of decision and confidence variables. For clarity, the presented likelihood function is cut at one value. Confidence is computed by inferring the posterior probability that the actor's action a was correct given the inferred state of the world d. The parameter $\rho\sigma_{act}\sigma_{conf}$ captures the correlation between the two Gaussian distributions. In this illustration, $\sigma_{act} = 1$, $\sigma_{conf} = 2.5$, and $\rho\sigma_{act}\sigma_{conf} = 0.6$

are equally probable. It is minimal when one of the categories is certain. The entropy of a bivariate and multivariate distribution is computed in a similar manner. Assume a set of categories generated by two variables x and y. The joint categories can be written as a matrix of their joint probabilities. The average entropy of this bivariate distribution is $H(x, y) = -\sum p(x, y) \log p(x, y)$. When variables x and y are independent, such that $p(x, y) = p(x) \times p(y)$, each cell in the joint probabilities matrix equals the cross product of the pertinent row and column marginal probabilities. In this orthogonal matrix, the joint entropy is maximal $H(x, y) = H(x, y)_{max}$. Such a matrix can also be generated when x and y are not independent, by multiplying the marginal cells, in the same manner by which expected cell proportions are computed in the *chi square* test. Values of $H(x, y) < H_{max}$ entail that the variables are correlated. It should be noted that H_{max} does not provide the maximal entropy for any given matrix irrespective of the marginals, because an equal distribution of probabilities always generates the largest entropy.

Another important quantity is called *mutual information*, $I(x; y)$, which conveys the amount of shared information between two variables. This amount reflects the reduction in joint entropy due to correlation relative to the maximal joint entropy when the variables are not correlated $I(x; y) = H(x, y)_{max} - H(x, y)$. When x and y are independent $I(x; y) = 0$. Mutual information can be also computed as $I(x; y) = H(x) + H(y) - H(x, y)$, since $H(x, y)_{max} = H(x) + H(y)$. The colon separating x and y indicates that the relations are symmetric $I(x; y) = I(y; x)$. Next, *conditional entropy* $H(x|y)$ or $H(y|x)$, reflects the average amount of entropy in one variable when the other variable is held constant. It can be shown that $I(x; y) = H(y) - H(y|x) = H(x) - H(x|y)$. Another useful formula expressing conditional entropy with three variables states that $I(x; y|z) = H(x|z) - H(x|y, z)$. That is, the mutual information between two variables conditioned on a third is equal to the entropy of the first conditioned on the third, minus the entropy of the first conditioned on the second and third variables.

The *Kullback–Leibler* (KL) divergence (Kullback, 1959), also known as *relative entropy* $D_{KL}(P \parallel Q)$ is a measure of how one probability distribution P is different from a second reference probability distribution Q. $D_{KL}(P \parallel Q) = \sum P(x) \log\left(\frac{P(x)}{Q(x)}\right)$, and $D_{KL}(Q \parallel P) = \sum P(x) \log\left(\frac{Q(x)}{P(x)}\right)$. The KL divergence between P and Q also expresses the

difference in entropies between the joint distribution P, Q and P. That is: $D_{KL}(P \parallel Q) = H(P, Q) - H(P)$. D_{KL} is always non-negative, and equals zero *if and only if* $P = Q$ (Feldman, 2021). The D_{KL} is an asymmetric measure of divergence, since $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$. A symmetric measure of divergence between two distributions P and Q is given by *Jeffreys divergence* (Jeffreys, 1992; Pearl, 1990). It is computed as the sum of two KL divergences $J(P, Q) = D_{KL}(P \parallel Q) + D_{KL}(Q \parallel P)$. It can be shown that $J(P, Q) = 2 \times H(P, Q) - H(P) - H(Q)$. The Jefferys divergence $J(P, Q)$ is always non-negative and equals zero *if and only if* $P = Q$.

Dayan (2023)'s *meta-$\mathcal{I}$*

Dayan (2023) has recently introduced an information-theoretic measure of metacognitive efficiency, termed *meta-$\mathcal{I}$*. Consider an actor who is engaged in a signal detection task with consequences of $a = 1$ or $a = -1$. The actor's response is r, and she is correct when $r = a$ and incorrect when $r \neq a$. Confidence in the actor's decision, c, is generated by a rater with partial information. The very same distributions (see Fig. 2B) that are used to calculate *meta-d'*, are used to assess the mutual information as a difference between two entropies, measuring how much information confidence conveys about decision accuracy.

$$meta - I = \quad H_2(r|a = -1) - H_2(r|c, a = -1) \tag{4}$$

where

$$H_2(r|a = -1) = \quad h_2[P(r|a = -1)] \tag{5}$$

and

$$H_2(r|c, a = -1) = \sum_c P(c|a = -1) h_2[P(r|c, a = -1)] \tag{6}$$

where

$$h_2[P(x)] = \sum_{x=0;1} -P(x) \log_2 P(x) \tag{7}$$

is the entropy of a Bernoulli random variable. The first entropy $H_2(r \mid a = -1)$, reflects the overall uncertainty about the actor's accuracy and is determined by d'. It ranges between 0 bits – when the actor is perfectly accurate and $d' \to \infty$ or systematically incorrect – to 1 bit when the actor's responses are uninformative about the stimulus ($d' \to 0$).

The second entropy, $H_2(r \mid c, a = -1)$, is the weighted average uncertainty about the residual accuracy after observing the confidence rating c, where the weights arrive from the probability of observing that rating c. The measure *meta-$\mathcal{I}$* is high if most of the uncertainty about the accuracy is removed by the confidence rating. In that case, the uncertainty remaining after the confidence rating $H_2(r \mid c, a = -1)$ is small, and *meta-$\mathcal{I}$* is large. The *meta-$\mathcal{I}$* has many desirable characteristics: (a) it is a relative measure, (b) accommodates to type 1 performance, (c) does not require any parametric assumptions, (d) scales with the number of confidence rating points, and (e) does not require the assumption of ratio or order of confidence ratings. One of the aims of the present study is to take advantage of this information-theoretic framework to develop complementary measures of metacognitive sensitivity.

Possible extensions of *meta-$\mathcal{I}$*

The *meta-$\mathcal{I}$* is a powerful measure of metacognitive efficiency. However, it has two features that limit its general applicability. First, it assumes that confidence judgments are based on the accuracy of the actor's response – an assumption that is not always corroborated by empirical support. For example, it has been shown (Koriat, 2008, 2012; Li & Ma, 2020) that people base their confidence on how common their choice is rather than on whether it is correct or incorrect. Second, *meta-$\mathcal{I}$* was specifically designed for two-choice tasks, and as such does not extend naturally to multi-alternative settings. To overcome these limitations, I offer a generalized index, *meta-$\mathcal{I}'$*, which does not depend on accuracy and can be applied to multi-alternative experimental settings. It captures the mutual information between the actor's response r and the stimulus-confidence variable c, s.

$$meta\text{-}\mathcal{I}' = I(r; c, s) \tag{8}$$

The *meta-$\mathcal{I}'$* equals zero bits when the actor's response r conveys no information about the joint structure of the rater's confidence ratings c and the stimuli s. It reaches its maximum when the actor's response is completely aligned (or perfectly correlated) with this structure. The upper bound of *meta-$\mathcal{I}'$* is determined by the variable with the lowest entropy, which is the one with the fewest discrete alternatives. For example, if stimulus is the variable with the lowest number of alternatives of four, then the upper bound is $\log_2(4) = 2$ bits. In the two-choice SDT task for which the original *meta-$\mathcal{I}$* was designed, the upper bound is $\log_2(2) = 1$ bits.

The measure *meta-$\mathcal{I}' = I(r; s, c)$* can be decomposed as the reduction in uncertainty about the actor's response after

accounting for both stimulus and confidence:

$$meta\text{-}\mathcal{I}' = I(r; s, c) = H(r) - H(r|s, c) \tag{9}$$

Alternatively, it can be expressed as the reduction in combined uncertainty of the actor's response and the joint stimulus-confidence variable:

$$meta\text{-}\mathcal{I}' = I(r; s, c) = H(r) + H(s, c) - H(r, s, c) \tag{10}$$

A useful related decomposition highlights how confidence reduces uncertainty in the actor's response beyond that explained by the stimulus alone:

$$meta\text{-}\mathcal{I}' = I(r; s, c) = H(r|s) - H(r|s, c) \tag{11}$$

These formulations clarify the information-theoretic meaning of $meta\text{-}\mathcal{I}'$ and enable its use in complex experimental designs with multiple stimuli, responses, and confidence levels – without relying on predefined correctness.

Metacognitive efficiency as divergence

I start by developing two straightforward measures of metacognitive efficiency, I call $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$. These measures are based on two classic information-theoretic divergence measures: the *Kullback-Leibler* (KL) (Kullback, 1959) and the *Jeffreys* (Jeffreys, 1992) divergence. These divergences are applied to the confidence-accuracy distributions $P(c = y \mid \text{Correct})$ and $Q(c = y \mid \text{Incorrect})$, which provide essential information regarding the ability of the rater to assess the correctness of the actor's decision. One salient feature of these discrete distributions is that the degree of overlap between the two is related to the rater's metacognitive sensitivity. When the rater is completely insensitive to the actor's decisions, the two distributions are indistinguishable. However, as the rater becomes more sensitive, the degree of separation between the two distributions increases. The KL divergence $D_{KL}(Q \parallel P)$ measures exactly this degree of separation in $bits$. The divergence between the correct and incorrect distributions amounts to $meta\text{-}\mathcal{KL}$:

$$\begin{aligned} meta\text{-}\mathcal{KL} &= D_{KL}(P \parallel Q) \\ &= \sum_y P(c = y \mid \text{Correct}) \log_2 \left(\frac{P(c = y \mid \text{Correct})}{Q(c = y \mid \text{Incorrect})} \right). \end{aligned} \tag{12}$$

When $meta\text{-}\mathcal{KL} = 0$, the rater cannot distinguish between the actor's correct and incorrect decisions. In that case, the two distributions are inseparable. As the rater becomes more sensitive to the actor's decisions, the divergence between the correct and incorrect distributions increases, and $meta\text{-}\mathcal{KL}$ increases proportionally, reflecting greater divergence between the confidence distributions.

In *Bayesian* terms, metacognitive efficiency can be interpreted as the information gained by a rater when updating her beliefs from a *prior* distribution Q (associated with incorrect decisions) to a *posterior* distribution P (associated with correct decisions). This gain is quantified in bits using the Kullback–Leibler (KL) divergence:

$$meta\text{-}\mathcal{KL} = D_{KL}(P \parallel Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}. \tag{13}$$

Equivalently, the KL divergence can be expressed as the difference between cross-entropy and entropy:

$$meta\text{-}\mathcal{KL} = H(P, Q) - H(P), \tag{14}$$

where

$$H(P) = -\sum_x P(x) \log_2 P(x) \tag{15}$$

is the Shannon entropy of P, and

$$H(P, Q) = -\sum_x P(x) \log_2 Q(x) \tag{16}$$

is the cross-entropy of P with respect to Q. Since cross-entropy is always greater than or equal to entropy, KL divergence is non-negative, with $D_{KL}(P \parallel Q) = 0$ if and only if $P = Q$. The quantity $meta\text{-}\mathcal{KL}$ thus measures the degree to which the confidence distribution for correct decisions differs from that for incorrect decisions. A higher value indicates greater metacognitive efficiency, reflecting the rater's ability to reduce uncertainty about the actor's correctness relative to a baseline defined by type 1 uncertainty. When $H(P) = 0$, confidence is perfectly concentrated on a single outcome (i.e., certainty), and $meta\text{-}\mathcal{KL}$ reaches its maximum. In this way, $meta\text{-}\mathcal{KL}$ serves as a relative measure of metacognitive efficiency, comparable to metrics such as the M-ratio or $meta\text{-}d' - d'_0$, as it tests type 2 sensitivity against the baseline of type 1 uncertainty. Although the reverse divergence $D_{KL}(Q \parallel P)$ can also be computed, it is less favorable in the context of metacognition, because here we wish to study how confidence follows the correct-response distribution.

The $meta\text{-}\mathcal{KL}$ divergence is not a metric, because it does not satisfy the triangle inequality (Cover & Thomas, 1991). It is also not symmetric in the two distributions, because $D_{KL}(P \parallel Q) \neq D_{KL}(Q \parallel P)$. A symmetric measure is

provided by $meta\text{-}\mathcal{J}$, which is based on the Jeffreys divergence (Jeffreys, 1992; Pearl, 1990):

$$\begin{aligned} meta\text{-}\mathcal{J} = J(P, Q) &= D_{KL}(P \parallel Q) + D_{KL}(Q \parallel P) \\ &= \sum_{y} P(c = y \mid \text{Correct}) \log_2 \left(\frac{P(c = y \mid \text{Correct})}{Q(c = y \mid \text{Incorrect})} \right) \\ &+ \sum_{y} Q(c = y \mid \text{Incorrect}) \log_2 \left(\frac{Q(c = y \mid \text{Incorrect})}{P(c = y \mid \text{Correct})} \right) \end{aligned} \tag{17}$$

$meta\text{-}\mathcal{J}$ operates in a very similar way to $meta\text{-}\mathcal{KL}$. When the rater is completely insensitive to the actor's decisions, $meta\text{-}\mathcal{J} = 0$. As the rater's metacognitive efficiency improves, the divergence grows.

This measure can be interpreted as the outcome of a reduction in the uncertainty in the combined confidence distributions for correct and incorrect decisions. That is, $meta\text{-}\mathcal{J}$ captures the extent to which the rater differentiates between correct and incorrect responses using confidence ratings. An equivalent entropy-based formulation makes this uncertainty-reduction interpretation more explicit:

$$\begin{aligned} meta\text{-}\mathcal{J} = J(P, Q) &= 2 \times H(P, Q) - H(P) - H(Q) \\ &= 2 \times H(c = y \mid \text{Correct},\ c = y \mid \text{Incorrect}) \\ &- H(c = y \mid \text{Correct}) - H(c = y \mid \text{Incorrect}) \end{aligned} \tag{18}$$

Cast in this way, two important characteristics of $meta\text{-}\mathcal{J}$ become apparent. First, the rater's goal is to minimize uncertainty in the actor's response by deducting uncertainty in the correct accuracy-confidence distribution, and/or in either the correct or incorrect confidence distributions. Second, these two sources contribute independently and equally to metacognitive efficiency. This is in contrast to the $meta\text{-}\mathcal{KL}$, which takes into consideration only the reduction of uncertainty in the correct accuracy-confidence distribution.

The structure of $meta\text{-}\mathcal{J}$, as revealed by Eq. 18, can explain a well-documented phenomenon called *the positive evidence bias* (Koizumi, Maniscalco, & Lau, 2015; Maniscalco, Peters, & Lau, 2016; Peters et al., 2017; Samaha, Barrett, Sheldon, LaRocque, & Postle, 2016; Zylberberg, Barttfeld, & Sigman, 2012). This bias refers to the rater's tendency to give more weight to evidence that is in favor of the actor's decision, while discounting evidence that is incongruent with the actor's decision. In terms of $meta\text{-}\mathcal{J}$, the positive evidence bias can be explained as resulting from the rater's disproportionate emphasis to reducing uncertainty in the correct accuracy-confidence distribution$H(c = y \mid \text{Correct})$, while downplaying – or even ignoring – the contribution of uncertainty in the incorrect distribution $H(c = y \mid \text{Incorrect})$. What $meta\text{-}\mathcal{J}$ reveals is that such an asymmetry in weighting positive and negative sources of evidence results in suboptimal metacognitive efficiency. An optimal level of metacognitive efficiency, as captured by $meta\text{-}\mathcal{J}$, is achievable only when both sources are equally weighted and effectively minimized by the rater.

While the divergence-based measures of metacognitive efficiency (e.g., meta-$\mathcal{KL}$, meta-$\mathcal{J}$) are defined here with respect to response accuracy, this is not a necessary requirement. These measures can be reformulated without explicit reference to correctness, relying instead on the way the actor internally partitions their responses. Such a label-free formulation allows one to assess the informativeness and structure of confidence judgments relative to the actor's own behavioral categories, enabling broader applicability in settings where external accuracy labels are unavailable, ambiguous, or theoretically inappropriate.

Initial validation of *meta-$\mathcal{KL}$* and *meta-$\mathcal{J}$*

I conducted an initial validation of the $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$ measures using the $meta\text{-}d'$ model (Maniscalco & Lau, 2012). Simulated accuracy-confidence distributions were generated with type 1 sensitivity fixed at $d'_0 = 2$, while type 2 sensitivity ($meta\text{-}d'$) varied across three levels: 1, 2, and 4. The number of rating buckets was held fixed at either four or eight. For each level of $meta\text{-}d'$, I computed the corresponding values of $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$. Figure 4 presents the results of these simulations. As can be noted, the degree of separability between the correct and incorrect accuracy-confidence distributions increased, both $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$ improved. For the four-point rating scale, when the M-ratio was 0.5, $meta\text{-}\mathcal{KL}$ equaled 0.36 and $meta\text{-}\mathcal{J}$ equaled 0.66. When the M-ratio was 1, $meta\text{-}\mathcal{KL}$ increased to 0.98 and $meta\text{-}\mathcal{J}$ to 1.76. When the M-ratio was 2, $meta\text{-}\mathcal{KL}$ reached 4.20 and $meta\text{-}\mathcal{J}$ reached 7.65. These results confirm that both divergence-based measures behave in a theoretically appropriate manner: they increase monotonically with metacognitive sensitivity, thereby supporting their validity as relative indices of metacognitive efficiency. Similar results were documented for the eight-point scale.

In another simulation, I examined the relationships between the SDT measures d'_0, $meta\text{-}d'$ and the information-theoretic measures $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$. The goal here is to provide support for their construct validity. As shown in Figs. 5A and B, the resulting heatmaps reveal lawful and interpretable relationships between the SDT and divergence-based metrics. Both $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$ increase monotonically with $meta\text{-}d'$, but critically, this increase is moderated by type 1 sensitivity (d'_0). At fixed metacognitive sensitivity, $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$ decrease as type 1 performance improves. These patterns reflect the desirable characteristics of a relative measure of metacognitive efficiency.

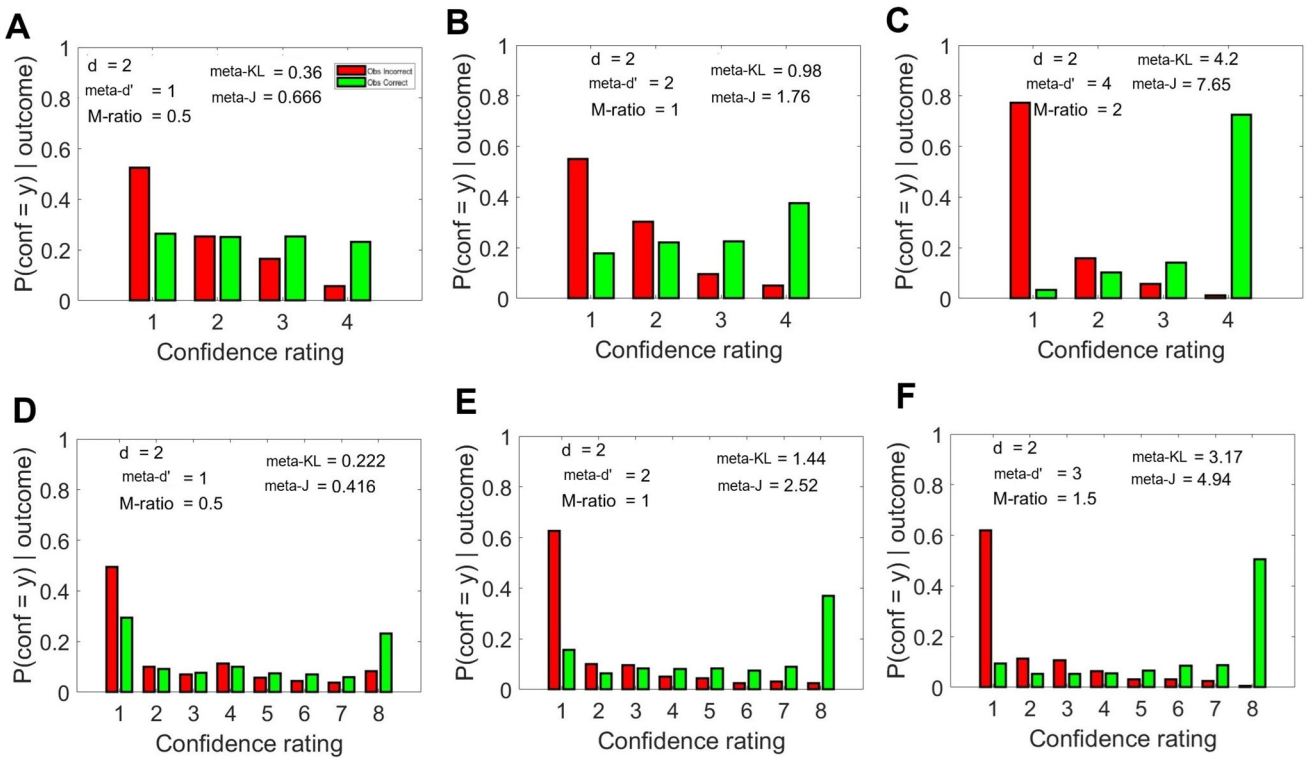


Fig. 4 Validation of *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$ using the *meta-*d' model. Panels **(A)**–**(C)** show simulations based on a four-point confidence rating scale; panels **(D)**–**(F)** use an eight-point scale. Each panel represents a different level of metacognitive sensitivity: hyposensitive (M-ratio $<$ 1, *left panels*), optimal (M-ratio = 1, *middle panels*), and hypersensitive (M-ratio $>$ 1, *right panels*). As metacognitive sensitivity increases, the separation between the correct and incorrect accuracy-confidence distributions also increases, resulting in higher values of both *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$

The same simulation data can be reanalyzed to explore the relationship between the M-ratio and the divergence measures. Because the M-ratio can span several orders of magnitude (e.g., very large when *meta-*$d' \gg d'_0$ and very small when *meta-*$d' \ll d'_0$), it serves as a useful benchmark for assessing the behavior of *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$ across a wide dynamic range. Figures 5C and D plot *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$ as a function of M-ratio. As can be noted, these divergence measures accommodate the different orders of magnitude very well. When the M-ratio is smaller than approximately 2, both measures increase steeply with M-ratio, while for larger values (e.g., M-ratio $>$ 5 or 10), the measures exhibit more gradual slopes. This pattern indicates that *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$ compensate for differences in type 1 d'_0, and therefore support their use as relative measures of metacognitive efficiency, alongside more traditional metrics such as meta-d'-d_0 or M-ratio.

Meta-$\mathcal{U}$: A multivariate dependence measure

A third measure of metacognitive efficiency – *meta-*$\mathcal{U}$ – is developed here. This measure is suitable for multi-alternative settings and does not rely on accuracy. To derive it, I adopt an information-theoretic technique known as *multivariate uncertainty analysis*, originally introduced by Garner (1962) and subsequently applied to a range of psychological questions (Garner & Morton, 1969; Garner & McGill, 1956; McGill, 1954; Ashby & Townsend, 1986; Fitousi, 2013). Although fully compatible with canonical information theory, this approach was specifically designed to model psychological constructs (Garner, 1962, 1974; Fitousi, 2013). As we will see, this framework enables the derivation of a metacognitive efficiency measure that captures the relationship between type 1 and type 2 performance in a particularly transparent manner. An analogous formulation within canonical information theory is also developed.

The goal is therefore to derive a measure, *meta-*$\mathcal{U}$, that evaluates how the rater's confidence ratings (c) reflect information about both the stimulus (s) and the actor's response (r). Unlike classical pairwise mutual information, this approach captures the full dependence structure among three variables, including higher-order interactions. One can visualize the joint distribution of stimulus, response, and confidence ratings, $P(s, r, c)$, as a three-dimensional contingency cube (Fig. 6), where each axis represents a variable

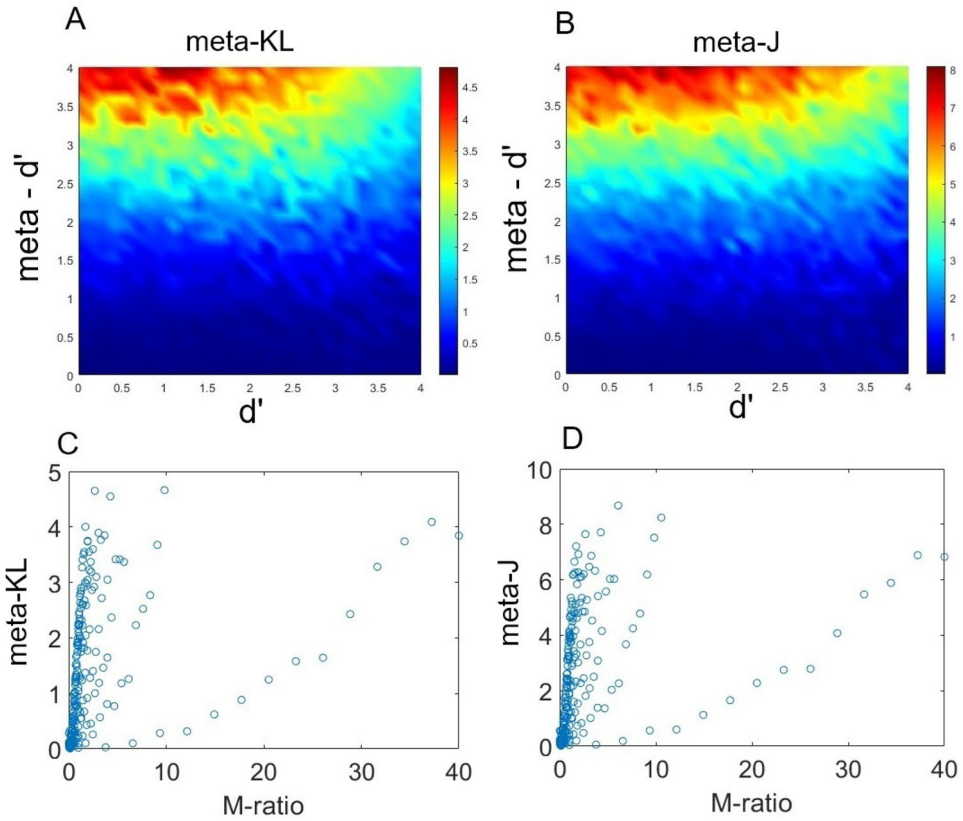


Fig. 5 **(A)** *meta-*$\mathcal{KL}$ as a function of d_0' and *meta-*d'. **(B)** *meta-*$\mathcal{J}$ as a function of d_0' and *meta-*d'. **(C)** *meta-*$\mathcal{KL}$ as a function of M-ratio (*meta-*d'/d_0'). **(D)** *meta-*$\mathcal{J}$ as a function of M-ratio. Simulations were conducted using the *meta-*d' model with a four-point confidence rating scale. Panels **(A)** and **(B)** are reproduced from Fitousi (2025b), published under the terms of the Creative Commons Attribution License (CC BY 4.0). © 2025 by the authors; published by MDPI, Basel, Switzerland

and each cell contains the joint probability of a specific (s, r, c) combination (see Attneave, 1959, p. 51). The cube reaches maximum entropy, $H_{\max} = H(s) + H(r) + H(c)$, when the variables are statistically independent. In this state, the distribution is maximally "spread out" and the cube is uniformly filled. This three-dimensional representation

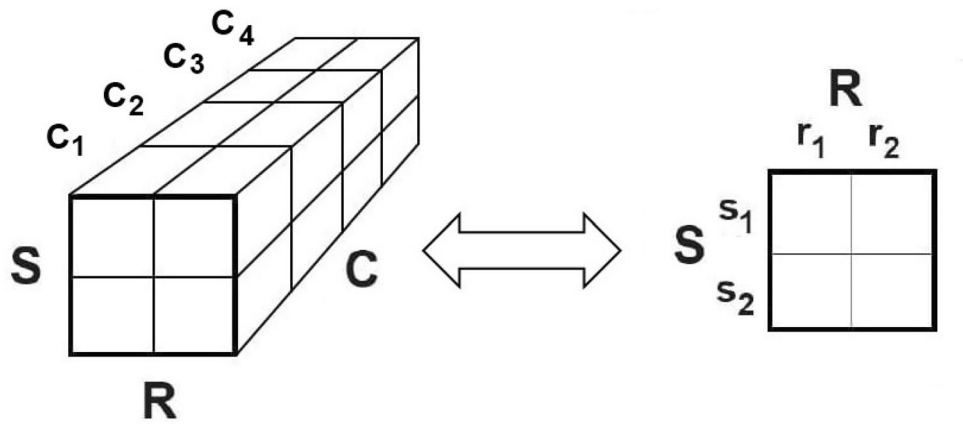


Fig. 6 *Left*: 3D matrix showing joint probabilities across stimulus S, response R, and confidence C. *Right*: 2D matrix of joint probabilities for S and R, collapsed over confidence. The 3D dependence is captured by $U(s : r : c)$, while the 2D projection reflects $U(s : r) = I(s; r)$. Their difference isolates the unique contribution of confidence

is broadly applicable, allowing the consideration of any full-factorial experimental design with arbitrary numbers of stimulus, response, and confidence rating bins.

The three-dimensional matrix is subjected to a total uncertainty constraint that can be partitioned onto various meaningful uncertainty components (Attneave, 1959; Garner & Hake, 1951; Garner, 1953, 1962; Garner & Morton, 1969; Fitousi, 2013). In his seminal book, Garner (1962) outlined a consistent and rigorous technique to accomplish this goal. Uncertainties are symbolized with the capital letter U (e.g., $U(r)$). The joint uncertainty of two variables is denoted $U(s,r)$, and of three variables $U(s,r,c)$. Garner defines the total multivariate dependence or *total constraint* in the systems as:

$$U(s:r:c) = H_{\max} - H(s,r,c), \tag{19}$$

which quantifies the deviation of the observed joint entropy from maximum entropy – that is, the structure in (s,r,c). This includes all interactions, not just pairwise. The pairwise dependence between s and r, which is also called *contingent uncertainty* is:

$$U(s:r) = H^{(2)}_{\max} - H(s,r) = I(s;r), \tag{20}$$

where $H^{(2)}_{\max}$ is the sum of the marginals of s and r. Note also that this quantity is analogous to the mutual information between stimuli and response, I(s;r), and reflects type 1 performance. To isolate the unique contribution of confidence, one can compute the *multiple contingent uncertainty*:

$$U(c:s,r) = U(s:r:c) - U(s:r), \tag{21}$$

which captures the additional dependence introduced by confidence beyond the stimulus-response association. The term U(c:s,r), which is essentially *meta-*$\mathcal{U}$, has an analogous identity in canonical information theory:

$$\textit{meta-}\mathcal{U} = U(c:s,r) = I(c;s,r). \tag{22}$$

The analogous term captures the mutual information between confidence and the stimulus-response variable. This identity, which is proven in Appendix A, is nontrivial: it shows that although $U(c:s,r)$ is conceptually grounded in Garner's multivariate uncertainty analysis, it aligns quantitatively with mutual information. Thus, the measure derived from Garner's framework has a direct analog in canonical information theory. Specifically, $U(c:s,r)$ (or equivalently, $I(c;s,r)$) quantifies how much additional information the confidence rating c conveys about the stimulus–response combination, beyond what would be expected by independence. The measure $U(c:s,r)$ can serve as a viable index of metacognitive efficiency, akin to *meta-*d'-d'_0 and the M-ratio. Like these measures, $U(c:s,r)$ isolates the informational contribution of confidence by subtracting the dependence already accounted for by the stimulus-response association. In this sense, it effectively corrects for type 1 performance. Furthermore, the mathematical equivalence $U(c:s,r) = I(c;s,r)$ enables a direct translation of this measure into canonical information-theoretic terms.

Interpretation of the *meta-*$\mathcal{U}$ index

The *meta-*$\mathcal{U}$ index is defined as the mutual information between confidence ratings and the joint stimulus-response pair. This measure reflects the extent to which confidence conveys information about both the stimulus and the actor's decision. It can be expressed as the reduction in uncertainty about confidence after observing the response, conditional on the stimulus:

$$\textit{meta-}\mathcal{U} = I(c;s,r) = H(c|s) - H(c|s,r) \tag{23}$$

This formulation emphasizes the informational gain provided by the actor's response. Alternatively, it can be expanded using entropy terms:

$$\textit{meta-}\mathcal{U} = I(c;s,r) = H(c) + H(s,r) - H(s,r,c) \tag{24}$$

This decomposition captures the overall structure of dependencies among stimulus, response, and confidence. Higher values of *meta-*$\mathcal{U}$ indicate that confidence judgments closely reflect both stimulus identity and behavioral response, signaling greater metacognitive efficiency. The maximum possible value of this index is constrained by the entropy of the variable with the fewest distinct categories – typically, the stimulus or response variables, which often involve only two alternatives.

It turns out, that *meta-*$\mathcal{U}$ conveys the same quantity as the *meta-*$\mathcal{I}'$ index proposed by Dayan (2023). This is due to the symmetry of mutual information:

$$\text{meta-}\mathcal{U} = \text{meta-}\mathcal{I}' \iff I(c;s,r) = I(r;c,s), \tag{25}$$

and in the case of two-choice SDT tasks, both amount quantitatively to Dayan's original meta-$\mathcal{I}$. Thus, meta-$\mathcal{U}$ functions both as a model and a metric for metacognitive efficiency. It is grounded in multivariate uncertainty analysis (Garner, 1962), and can be applied to any number of stimuli, responses, and confidence ratings.

Validation of *meta-$\mathcal{U}$*

Our novel measure, *meta-$\mathcal{U}$*, is a genuine index of metacognitive efficiency. To establish its validity relative to canonical signal detection theory (SDT) metrics, I tested whether *meta-$\mathcal{U}$* and its components $U(s:r)$, $U(s:r:c)$ systematically covary with SDT indices: d'_0, *meta-d′*, and metacognitive efficiency *M-ratio*. Using simulations based on the *meta-d′* framework (Maniscalco & Lau, 2012), confidence-accuracy distributions were generated by orthogonally varying type 1 sensitivity (d'_0) and type 2 sensitivity (*meta-d′*). For each pair, *meta-$\mathcal{U}$* was computed.

Figure 7A shows a heatmap with d'_0 on the x-axis, *meta-d′* on the y-axis, and *meta-$\mathcal{U}$* as color. *meta-$\mathcal{U}$* rises with *meta-d′*, but this increase is attenuated as d'_0 grows, reflecting its desirable relative scaling: at fixed metacognitive sensitivity, *meta-$\mathcal{U}$* decreases as type 1 performance improves. This mirrors the logic behind relative metacognitive efficiency in indices like *meta-d′* $-d'_0$ and *M-ratio*. Figure 7B plots *meta-$\mathcal{U}$* against *M-ratio* $=$ *meta-d′*$/d'_0$, showing a monotonic relation. Crucially, *meta-$\mathcal{U}$* is sensitive across magnitudes of *M-ratio*, supporting its construct validity as a robust, general-purpose metacognitive efficiency measure.

A strong correspondence between *meta-$\mathcal{U}$*, d'_0, *meta-d′*, and *M-ratio* is expected. More surprisingly, lawful relations emerge between components of *meta-$\mathcal{U}$*, namely $U(s:r)$ and $U(s:r:c)$, and those of *M-ratio*, namely d'_0 and *meta-d′*. This remarkable correspondence suggests the two measures operate under similar principles. Specifically, $U(s:r)$, indexing type 1 sensitivity, is positively and monotonically related to d'_0, while $U(s:r:c)$, capturing total system constraint, relates positively and monotonically to both *meta-d′* and d'_0. The left column of Fig. 8 shows simulations where $U(s:r)$ is plotted as a function of d'_0 for fixed *meta-d′* $= 1, 2, 3, 4$. The results reveal a consistent monotonic increase of $U(s:r)$ with d'_0, independent of *meta-d′*. The right column presents $U(s:r:c)$ as a function of *meta-d′* for fixed $d'_0 = 1, 2, 3, 4$. Here, $U(s:r:c)$ increases monotonically with *meta-d′*, but the slope diminishes as d'_0 grows, becoming nearly flat at $d'_0 = 4$. This pattern reflects the constraint that the correlation between two variables limits their joint correlation with a third variable. As type 1 performance becomes more efficient, there is less room to improve combined type 1 and type 2 performance. This explains how *meta-$\mathcal{U}$* compensates for the rater's sensitivity by offsetting the information transmitted by the actor, $U(s:r)$.

Information-theoretic foundations of *meta-$\mathcal{U}$*

Several researchers (Bowman et al., 2022; Killeen & Taylor, 2001) have proposed that type 1 discriminability in SDT tasks can be quantified as the amount of mutual information shared between stimuli and response $I(s;r)$. This is equivalent to the measure $U(s:r)$, given the identity $I(s;r) = U(s:r)$. Foundational work in cognitive science already noted the deep connection between discriminability and mutual information (Garner, 1962; Miller, 1956). For example, in his seminal paper on the capacity limitations of short-term memory, Miller (1956) measured the amount of information, T, between stimulus and response in an absolute

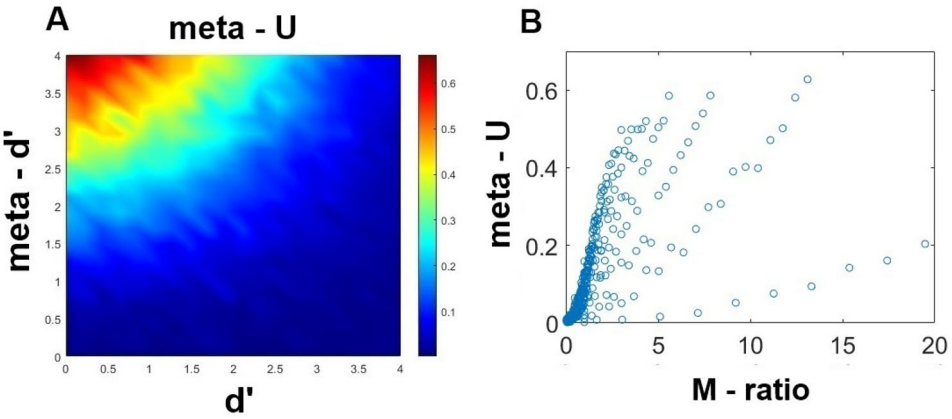


Fig. 7 Validation of ***meta-$\mathcal{U}$*** using the ***meta-d′*** model. Simulations assumed four confidence levels. Both type 1 sensitivity (d') and type 2 sensitivity (*meta-d′*) were varied orthogonally. (A) *meta-$\mathcal{U}$* as a joint function of d' (x-axis) and *meta-d′* (y-axis). **(B)** *meta-$\mathcal{U}$* plotted as a function of metacognitive efficiency (*M-ratio* $=$ *meta-d′*$/d'$). The results demonstrate that *meta-$\mathcal{U}$* increases with metacognitive sensitivity, but in a manner that is modulated by type 1 performance – consistent with its role as a relative (rather than absolute) measure of metacognitive efficiency. Panel A of the figure was reproduced from Fitousi (2025b), published under the terms of the Creative Commons Attribution License (CC BY 4.0). © 2025 by the authors; published by MDPI, Basel, Switzerland

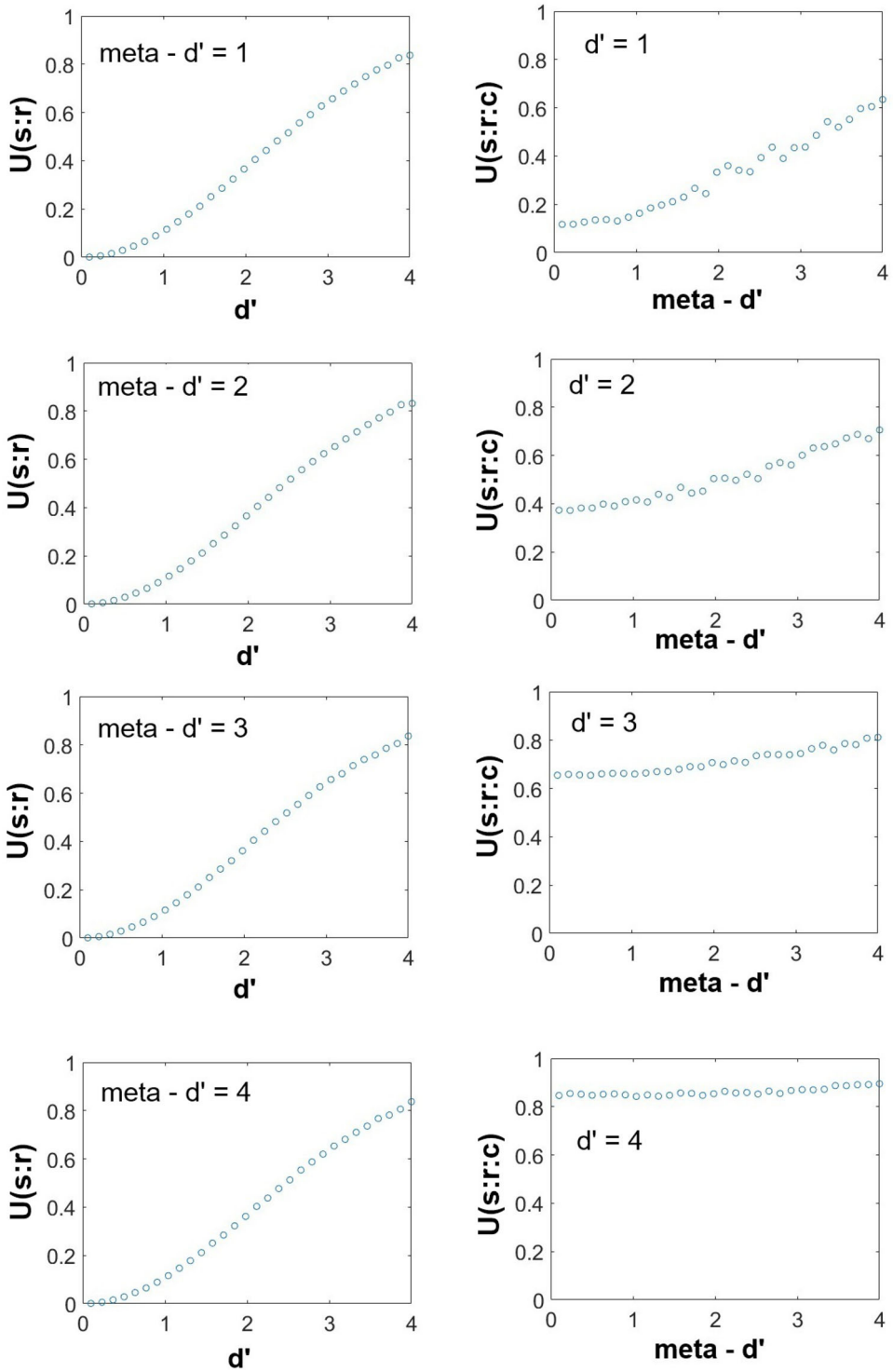


Fig. 8 Validation of *meta-𝒰* and its components $U(s:r)$ and $U(s:r:c)$ against the *meta-d′* model parameters d_0' and *meta-d′*. *Left column:* $U(s:r)$ as a function of d_0' for *meta-d′* $= 1, 2, 3, 4$, showing a monotonic relationship that captures type 1 sensitivity. *Right column:* $U(s:r:c)$ as a function of *meta-d′* for $d_0' = 1, 2, 3, 4$, indicating a monotonic relationship with *meta-d′* modulated by d_0'. Together, these patterns explain how *meta-𝒰* $= U(s:r:c) - U(s:r)$ reflects metacognitive efficiency analogously to *meta-d′* $- d_0'$

identification task to estimate the effective capacity of participants. Garner (1953) had deployed a comparable measure, $U(s : r)$, under the name *contingent uncertainty* (see for Algom & Fitousi, 2016). In rating studies (Garner, 1960; Hake & Garner, 1951), participants evaluated the legibility of handwriting samples across varying scale granularities. These studies revealed an upper bound of approximately 3.3 bits in the information transmitted between stimulus and rating response.

Over the decades, mutual information – under different notations such as T, $I(s;r)$, or $U(s : r)$ – has been studied extensively (McGill, 1954; Miller, 1955; Killeen & Taylor, 2001). Statistically, it is known to follow a chi-square distribution: $knT \approx \chi^2$, where n is the number of observations and $k = 2\ln 2 \approx 1.386$ (McGill, 1954). Miller (1955) proposed an unbiased estimator for T: $\hat{T} = T - \frac{(r-1)(c-1)}{k}$, where r and c are the numbers of stimulus and response categories. Killeen and Taylor (2001) further showed that T approximates about 10% of $d_0'^2$. In our context, $U(s : r)$ represents the reduction in uncertainty about s after observing r – a direct information-theoretic analog of type 1 sensitivity.

The relations between information theory and SDT quantities can be further explicated by the construction of *isoinformation ROCs curves* (Killeen & Taylor, 2001). These curves are analogous to the ROC curves (shown in Fig. 9A) generated in type 1 SDT (Macmillan & Creelman, 2004). The isoinformation ROCs curves depict the trade-off between hit rate and false alarm as a function of the decision criterion. Figure 9B presents the amount of information transmitted from stimuli to responses, $U(s : r)$ in a similar hit–false alarm space. The contours through which these curves are drawn conserve the uncertainty reduced (or information transmitted) by the actor relative to the stimuli. These contours are very similar to those generated by Pollack and Norman (1964) for their non-parametric measure A'. Isoinformation ROC curves can also be drawn for performance in the type 1 and type 2 tasks relative to that in the type 1 task. We simply vary $U(s : r)$ and $U(s : r : c)$, and computing $meta\text{-}\mathcal{U} = U(s : r : c) - U(s : r)$ at each point. Figure 9C presents those lines. As can be noted, the lines conserve metacognitive efficiency across changes in type 1 and type 2 sensitivity, giving currency to the desirable property of $meta\text{-}\mathcal{U}$ as a relative measure.

Validation with the second-order model

In the next set of simulations, I harnessed the second-order model (Jang et al., 2012; Fleming & Daw, 2017; Schulz et al., 2023), which offers a more sophisticated framework than the $meta\text{-}d'$ model (Maniscalco & Lau, 2012; Fleming, 2017). This model assumes that the actor and rater are sampling evidence from two separate distributions. The parameters σ_{act} and σ_{conf} determine the noise levels in the actor's and rater's evidence distributions, respectively. Increasing or decreasing the rater's distribution noise relative to that of the actor's reduces or increases metacognitive sensitivity. The parameter ρ controls for the correlation between the distributions. Confidence ratings are generated by partitioning the rater's posterior probability space (i.e., 0–1) to into n segments, where n corresponds to the number of rating points. The segments do not need to be equal in size. Different partitions

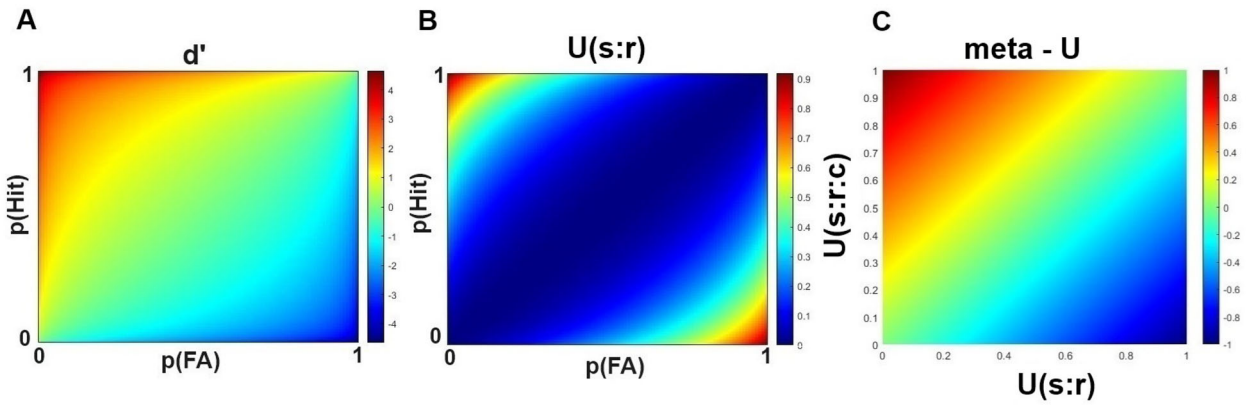


Fig. 9 **(A)** Heatmap of the standard SDT discriminability index d_0' as a function of hit rate p(Hit) and false-alarm rate p(FA) for type 1 performance. **(B)** Heatmap of mutual information $U(s{:}r)$ between stimuli and response across values of p(Hit) and p(FA), with contour lines indicating constant transmitted information (isoinformation ROC curves). **(C)** Heatmap of metacognitive efficiency $meta\text{-}\mathcal{U} = U(s{:}r{:}c) - U(s{:}r)$ as a function of type 1 sensitivity $U(s{:}r)$ and multivariate dependence $U(s{:}r{:}c)$, with contours connecting points of equivalent relative metacognitive efficiency (isoinformation meta-ROC curves)

may lead to different quantities of transmitted information (Dayan, 2023), but once the experimenter is consistent with a given partition, she can meaningfully compare the observed information values across participants. The partition of the rater's posterior probability (along with the number of rating scales) can be used by the experimenter to administer rating scales that maximize information transmission (Bang et al., 2017).

As can be noted in Fig. 10(A)–(C), when $\sigma_{\text{conf}} \gg \sigma_{\text{act}}$, the M-ratio is approximately 0.3, and the information measures $meta\text{-}\mathcal{KL}$, $meta\text{-}\mathcal{J}$, and $meta\text{-}\mathcal{U}$ yield relatively low values. For example, with a four-point rating scale, $meta\text{-}\mathcal{KL} = 0.13$, $meta\text{-}\mathcal{J} = 0.28$, and $meta\text{-}\mathcal{U} = 0.02$. Comparable results are obtained with an eight-point rating scale. In both cases, correct and incorrect confidence-accuracy distributions exhibit considerable overlap.

In Fig. 10(D)–(F), when the noise levels in the actor's and rater's distributions are equal ($\sigma_{\text{conf}} = \sigma_{\text{act}}$), the resulting M-ratio is 1.39, and the information measures yield intermediate values, with moderate overlap between the confidence–accuracy distributions. For example, with a four-point rating scale, $meta\text{-}\mathcal{KL} = 1.73$, $meta\text{-}\mathcal{J} = 3.10$, and $meta\text{-}\mathcal{U} = 0.19$. Comparable results are obtained with the eight-point scale. In both cases, correct and incor-

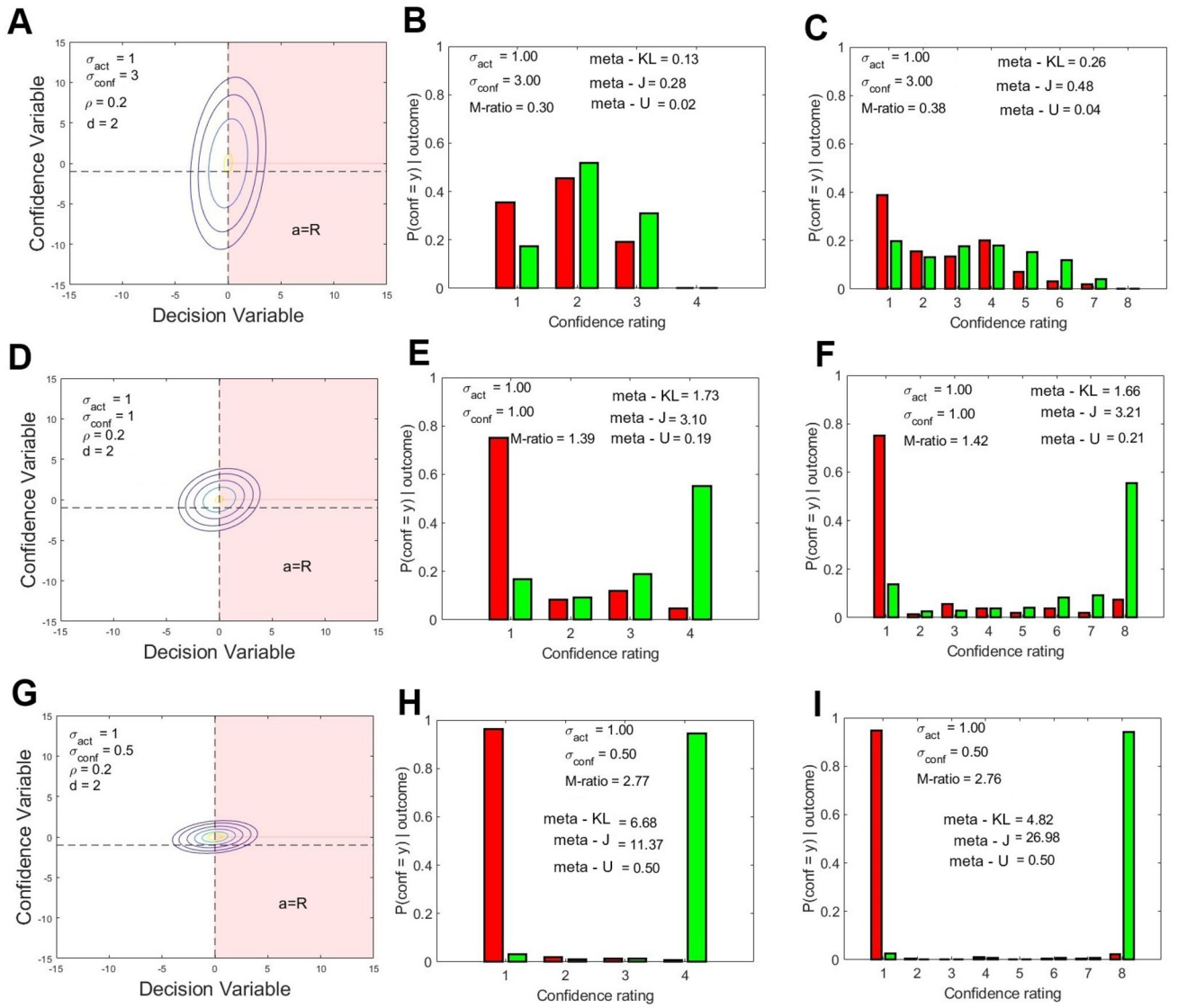


Fig. 10 **(A)–(C)**: Second-order model with $\sigma_{\text{conf}} \gg \sigma_{\text{act}}$. **(D)–(F)**: Second-order model with $\sigma_{\text{conf}} = \sigma_{\text{act}}$. **(G)–(I)**: Second-order model with $\sigma_{\text{conf}} < \sigma_{\text{act}}$. In each triplet of panels, the *left panel* shows the second-order model representation; the *middle panel* displays the confidence–accuracy distributions for a four-point scale; and the *right panel* displays the distributions for an eight-point scale. Posterior probability partition for the four-point scale: 0–0.8, 0.8–0.875, 0.875–0.95, and 0.95–1. Parameters: $\rho = 0.2$, $d' = 2$. Each simulation sampled 1000 trials. Bars represent the confidence–accuracy distributions for correct and incorrect responses

rect confidence–accuracy distributions exhibit intermediate degree of overlap.

Finally, in Fig. 10(G)–(I), when the actor's distribution is noisier than the rater's ($\sigma_{conf} < \sigma_{act}$), the M-ratio rises to 2.77, the information measures become very large, and overlap between confidence–accuracy distributions is minimal. For instance, with a four-point rating scale, $meta\text{-}\mathcal{KL} = 8.68$, $meta\text{-}\mathcal{J} = 11.37$, and $meta\text{-}\mathcal{U} = 0.50$. Comparable results are observed with the eight-point scale.

Convergence with measures of divergence

Another way of validating the novel information measures advanced here is by assessing their relatedness. If these measures gauge the same construct of metacognitive efficiency, they should correlate. To examine this hypothesis, I performed the following simulations. I fixed the value of σ_{act} at 1, while incrementally increasing the value of σ_{conf} from 0.3 to 3 creating 100 equally spaced values. All other parameters were held constant ($\rho = 0.2$). I then computed the corresponding values of $meta\text{-}\mathcal{KL}$, $meta\text{-}\mathcal{J}$, and $meta\text{-}\mathcal{U}$, and plotted them against one another. Figure 11 presents the results. As can be noticed, the three measures exhibit remarkable positive, almost linear correspondence. The strong correspondence between $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$ is not really surprising, given their complimentary mathematical relations, but their correspondence with the mutual information measure, $meta\text{-}\mathcal{U}$, is indeed remarkable, especially given the fact that they are designed to capture distinct aspects of information transmission. These high correlations provide more support in their construct validity.

Converging validation: Information transmitted between actor and rater

I capitalize on a powerful, non-trivial prediction of the second-order model (Fleming & Daw, 2017; Jang et al., 2012) regarding the influence of its ρ parameter on metacognitive efficiency, showing how the same model predictions can be captured in terms of the present information-theory model. As noted earlier, the second-order model (Fleming & Daw, 2017; Jang et al., 2012) treats the actor and rater as two independent entities, but also allows them to share information. The evidence sampled by the rater can be made contingent on the evidence adduced by the actor, via the ρ parameter. This parameter captures the degree of correlation between the two Gaussian distributions. Fleming and Daw (2017) have shown that, as the ρ parameter is increased, and the rater and actor share more evidence, the rater's metacognitive efficiency decreases. This is a rather surprising prediction. From a Bayesian point of view, this phenomenon can be explained as resulting from a bias in the rater's prior toward the actor's prior, which leads to a rater's posterior probability that is less accurate compared to an unbiased prior.

Can our information theory model account for this phenomenon? From an information-theoretic vantage point, the rater and actor share information, which places an upper bound on the rater's independent mutual information given the actor's response. Kullback (1959) (see Eq. 4.3, p.8), proved that the mutual information between two univariate Gaussian distributions $g(x)$ and $g(y)$, composing a bivariate distribution $f(X, Y)$, depends solely on the correlation

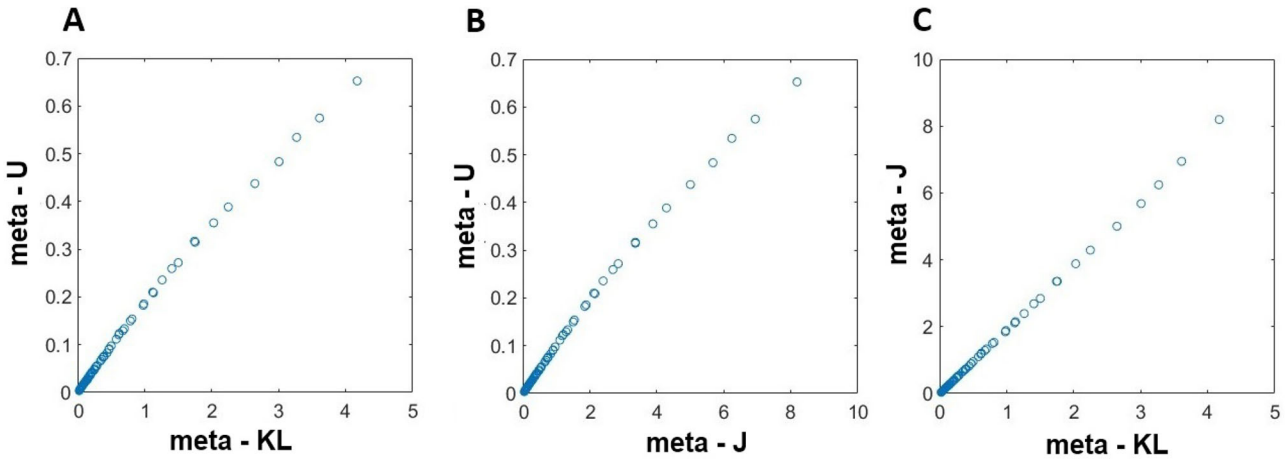


Fig. 11 Relationships between the mutual information measure $meta\text{-}\mathcal{U}$ and the divergence-based measures $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$. **(A)** Correspondence between $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{U}$; **(B)** correspondence between $meta\text{-}\mathcal{J}$ and $meta\text{-}\mathcal{U}$; **(C)** correspondence between $meta\text{-}\mathcal{KL}$ and $meta\text{-}\mathcal{J}$. Simulations were conducted using the second-order model. The actor's noise parameter was set to $\sigma_{act} = 1$, while the rater's noise parameter σ_{conf} was varied incrementally from 0.3 to 3. The figure reproduced from Fitousi (2025b), published under the terms of the Creative Commons Attribution License (CC BY 4.0). © 2025 by the authors; published by MDPI, Basel, Switzerland

parameter ρ, and is given by

$$I(X;Y) = -\frac{1}{2}\log\left(1-\rho^2\right) \tag{26}$$

Assuming the actor samples from a Gaussian distribution $g(s_a)$, and the rater samples from $g(s_c)$, correlated by the parameter $\rho_{s_a s_c}$, their joint distribution can be described by the bivariate Gaussian $f(s_a, s_c)$. The information exchanged between the actor and the rater is thus:

$$I(s_a;s_c) = -\frac{1}{2}\log\left(1-\rho^2_{s_a s_c}\right) \tag{27}$$

Figure 12 illustrates three levels of ρ and the resulting mutual information $I(s_a; s_c; r)$ between the actor's evidence distribution, the rater's evidence distribution, and the actor's response. As ρ increases, this quantity increases accordingly. When $\rho = 0$, no information is exchanged, and $I(s_a; s_c; r) = 0$. For moderate correlation ($\rho = 0.6$), mutual information is approximately 0.32; for high correlation ($\rho = 0.9$), it rises to approximately 1.2.

It thus becomes apparent that the maximal amount of information the rater can extract from the actor's response – practically corresponding to her metacognitive efficiency, *meta-$\mathcal{U}$* – is bounded from above by the quantity $I(s_a; r) - I(s_c; r)$. This expression captures the effective shared information between stimulus and response, after removing the portion of mutual information attributable to correlation. Importantly, this resulting quantity is a decreasing function of $\rho_{s_a s_c}$. Because the divergence-based measures *meta-$\mathcal{KL}$* and *meta-$\mathcal{J}$* are strongly correlated with *meta-$\mathcal{U}$*, they are likewise expected to decrease with increasing $\rho_{s_a s_c}$. Figure 13 presents simulation results in which the $\rho_{s_a s_c}$ parameter was incrementally increased, and the SDT-based measure M-ratio, along with the information-theoretic metrics *meta-$\mathcal{U}$*, *meta-$\mathcal{KL}$*, and *meta-$\mathcal{J}$*, were computed.

Figure 13A shows the expected decrease in M-ratio as a function of $\rho_{s_a s_c}$, replicating the finding by Fleming and Daw (2017). Figure 13B reveals that $U(s{:}r)$ – a measure of type 1 sensitivity – remains unaffected by $\rho_{s_a s_c}$, as expected. In contrast, Fig. 13C shows that $U(s{:}r{:}c)$, which reflects total uncertainty in the system (including both type 1 and type 2 sensitivity), decreases with increasing $\rho_{s_a s_c}$, consistent with the monotonic decline in *meta-$\mathcal{U}$* depicted in Fig. 13D. Finally, Figs. 13E and F confirm that the divergence-based measures of metacognitive efficiency exhibit a similar decreasing trend.

All these findings lead to a novel prediction: the mutual information $I(s_a; s_c; r)$ between actor's and rater's distributions and actor's response is expected to be an increasing, rather than a decreasing, monotonic function of $\rho_{s_a s_c}$. This is because this quantity captures the portion of mutual information between stimuli and response due to redundancy, or correlation. Figure 14(A)–(B) presents the results of a simulation in which $\rho_{s_a s_c}$ was increased incrementally in small steps, while the corresponding values of $I(s_a; s_c)$ and $I(s_a; s_c; r)$ were recorded. Figure 14A illustrates the relationship between mutual information $I(s_a; s_c)$ and $\rho_{s_a s_c}$, formally defined in Eq. 27, which shows a positive monotonic relationship. Figure 14B plots the triple mutual information $I(s_a; s_c; r)$ as a function of $\rho_{s_a s_c}$. As can be noted, the prediction was well supported. These results show that an information-theory approach to metacognition can provide a detailed, transparent, and quantitatively exact model of the exchange of information between rater, actor, and experimenter. Such a model can elucidate the sources of uncertainty and their impact on metacognitive efficiency, an account that

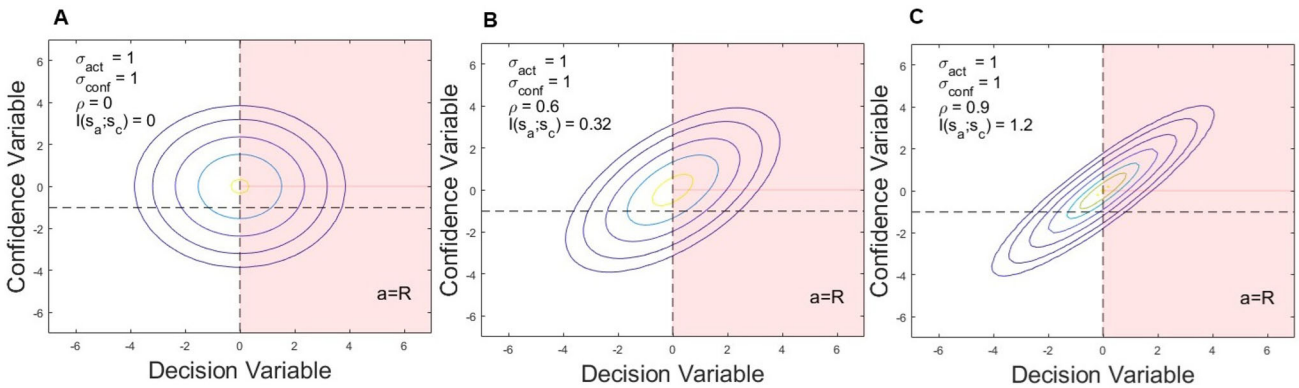


Fig. 12 Relationship between the second-order model parameter $\rho_{s_a s_c}$, which reflects the correlation between the actor's and rater's evidence distributions, and $I(s_a; s_c; r)$, the amount of information exchanged between actor and rater. **(A)** When $\rho_{s_a s_c} = 0$, no information is exchanged: $I(s_a; s_c; r) = 0$. **(B)** For a moderate correlation ($\rho_{s_a s_c} = 0.6$), the exchanged information is moderate: $I(s_a; s_c; r) = 0.32$. **(C)** For a high correlation ($\rho_{s_a s_c} = 0.9$), the information exchanged is relatively high: $I(s_a; s_c; r) = 1.2$

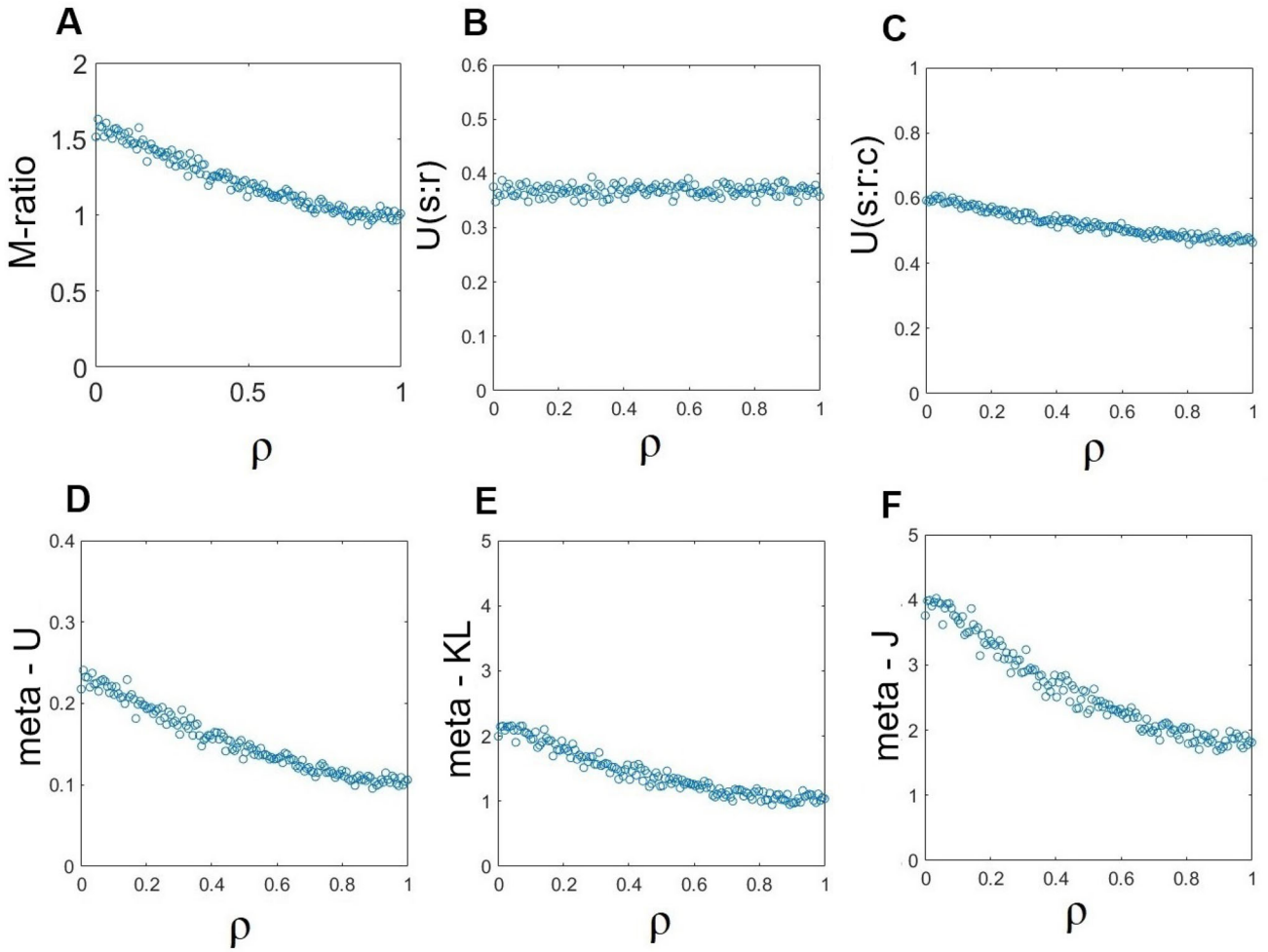


Fig. 13 Simulations with the second-order model showing the relationships between the parameter ρ and various SDT and information-theoretic quantities. **(A)** M-ratio as a function of ρ. **(B)** $U(s{:}r)$ as a function of ρ. **(C)** $U(s{:}r{:}c)$ as a function of ρ. **(D)** $meta\text{-}\mathcal{U}$ as a function of ρ. **(E)** $meta\text{-}\mathcal{KL}$ as a function of ρ. **(F)** $meta\text{-}\mathcal{J}$ as a function of ρ

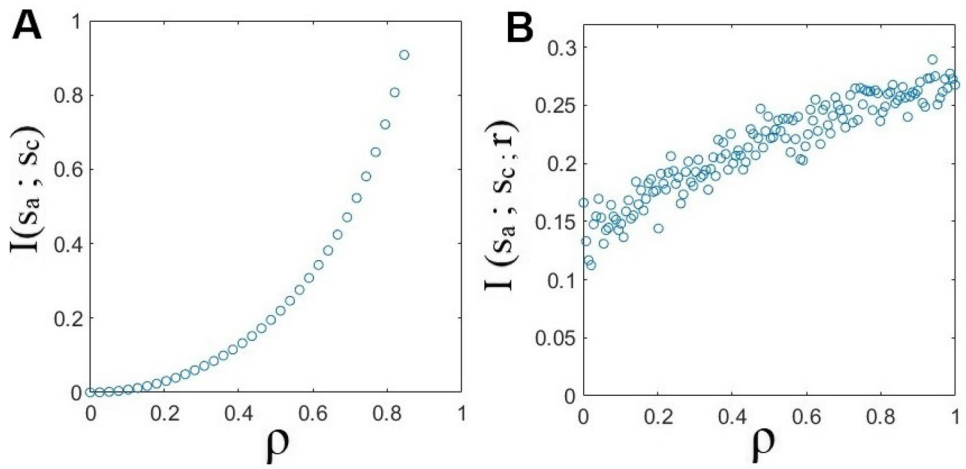


Fig. 14 **(A)** Mutual information $I(s_a; s_c)$ shared between the actor's $g(s_a)$ and the rater's $g(s_c)$ distributions. **(B)** Triple mutual information $I(s_a; s_c; r)$ as a function of the correlation parameter $\rho_{s_a s_c}$

is not readily available by other, more popular approaches (e.g., SDT).

Effect of number of ratings scales

A desirable property of information measures is their tendency to scale with the number of confidence-rating buckets (Garner, 1960; Killeen & Taylor, 2001; Dayan, 2023). As the number of rating buckets increases, the rater can be more sensitive because she can resolve greater amounts of uncertainty regarding the actor's decisions, and as a result, transmit more information to the experimenter. SDT measures, such as M-ratio, are not expected to exhibit this scaling behavior (Dayan, 2023). In this set of simulations, I sought to assess whether- and under what- conditions do our novel information-theory measures of metacognitive efficiency scale with the number of confidence-rating alternatives. The simulations were held with the second-order model (Fleming & Daw, 2017). In all simulations, the ρ parameter was set to 0, to allow for maximal metacognitive sensitivity possible, and the number of trials simulated was 1000. In the first simulation, I tested for situations in which the rater is metacognitively inefficient, by setting $\sigma_{\text{conf}} \gg \sigma_{\text{act}}$. I then increased the number of confidence-rating buckets incrementally from the minimum possible of two to twenty. The range of accuracy covered by each of the inner rating buckets was determined by dividing the posterior accuracy of the rater in the range 0.8-0.95 into the pertinent number of rating buckets. The remaining two (outer) buckets consisted of the values 0–0.8, and 0.95—1. I then computed the corresponding values of M-ratio, *meta-*$\mathcal{U}$, *meta-*$\mathcal{KL}$, and *meta-*$\mathcal{J}$ for each number of rating-buckets.

Figure 15A–D presents the results. As can be noted, the information-theory measures increased substantially as the number of rating buckets increased. The M-ratio exhibited a small increase as well, but not comparable to the information measures. In the second and third simulations, I tested conditions in which the rater is either perfectly sensitive or hyper-sensitive (Maniscalco & Lau, 2012, 2014; Fleming, 2017), by setting $\sigma_{\text{conf}} = \sigma_{\text{act}}$ and $\sigma_{\text{conf}} < \sigma_{\text{act}}$, respectively. Figure 15E–I shows the results. Although the information-theoretic measures scaled with the number of rating buckets to some extent, this scaling was much less pronounced compared to Simulation 1. The M-ratio, in particular, did not exhibit any noticeable increase.

Taken together, these results suggest that when the rater's efficiency is sub-optimal (Shekhar & Rahnev, 2021b), the information-theory measures of metacognitive sensitivity scale with the number of rating buckets, as expected. This property seems less salient when the rater is maximally sensitive, or even hyper-sensitive. Why is it so? One explanation is provided by information theory itself: the rater's performance is bounded from above by the information available in the actor's response. As long as the rater is extracting less information from the actor's behavior than is available by the upper bound, there is still room for improvement in the rater's sensitivity as a function of the number of rating-buckets. However, once this upper bound is reached, the rater cannot improve her sensitivity beyond what is dictated by that upper bound.

Empirical validation

While the present paper is theoretical in nature, it would be fair to say that there remains considerable uncertainty about which models of metacognition are best. It is now widely acknowledged that the meta-d′ model is not a process model (but see, Rausch, Hellmann, & Zehetleitner, 2023). Furthermore, the model proposed by Fleming and Daw (2017) performed rather poorly in a recent model comparison study (Shekhar & Rahnev, 2024). One could argue that a more fruitful strategy for validating metacognitive measures is to evaluate their empirical properties using real behavioral data (see for example, Rahnev, 2025). This is precisely what a recent study by Fitousi (2025b) aimed to do. The paper examined the core information-theoretic constructs introduced here – *meta-*$\mathcal{U}$, *meta-*$\mathcal{KL}$, and *meta-*$\mathcal{J}$ – using data from two face-matching experiments (Fitousi, 2024). In Experiment 1, participants completed the Glasgow Face Matching Task (GFMT; Burton, White, & McNeill, 2010), and in Experiment 2, they performed the Kent Face Matching Task (KFMT; Fysh & Bindemann, 2018). In both experiments, participants judged whether images of face pairs depicted the same or different identities, and then rated their confidence on a four-point scale. This design enabled the computation of both traditional metacognitive indices (e.g., M-ratio) and the proposed information-theoretic measures.

The results revealed strong, systematic relationships between these indices, providing empirical support for several theoretical claims. Figure 16 presents representative findings from Experiment 2, reproduced from Fitousi (2025b) under the terms of the Creative Commons Attribution License (CC BY 4.0). This alignment between theoretical predictions and observed data strengthens the case for the construct validity and explanatory power of the proposed framework.

General discussion

Dayan (2023) has recently proposed–meta-$\mathcal{I}$ – an information-theoretic measure of metacognitive efficiency. The present work extends and complements this approach by introducing three novel measures of metacognitive efficiency: two diver-

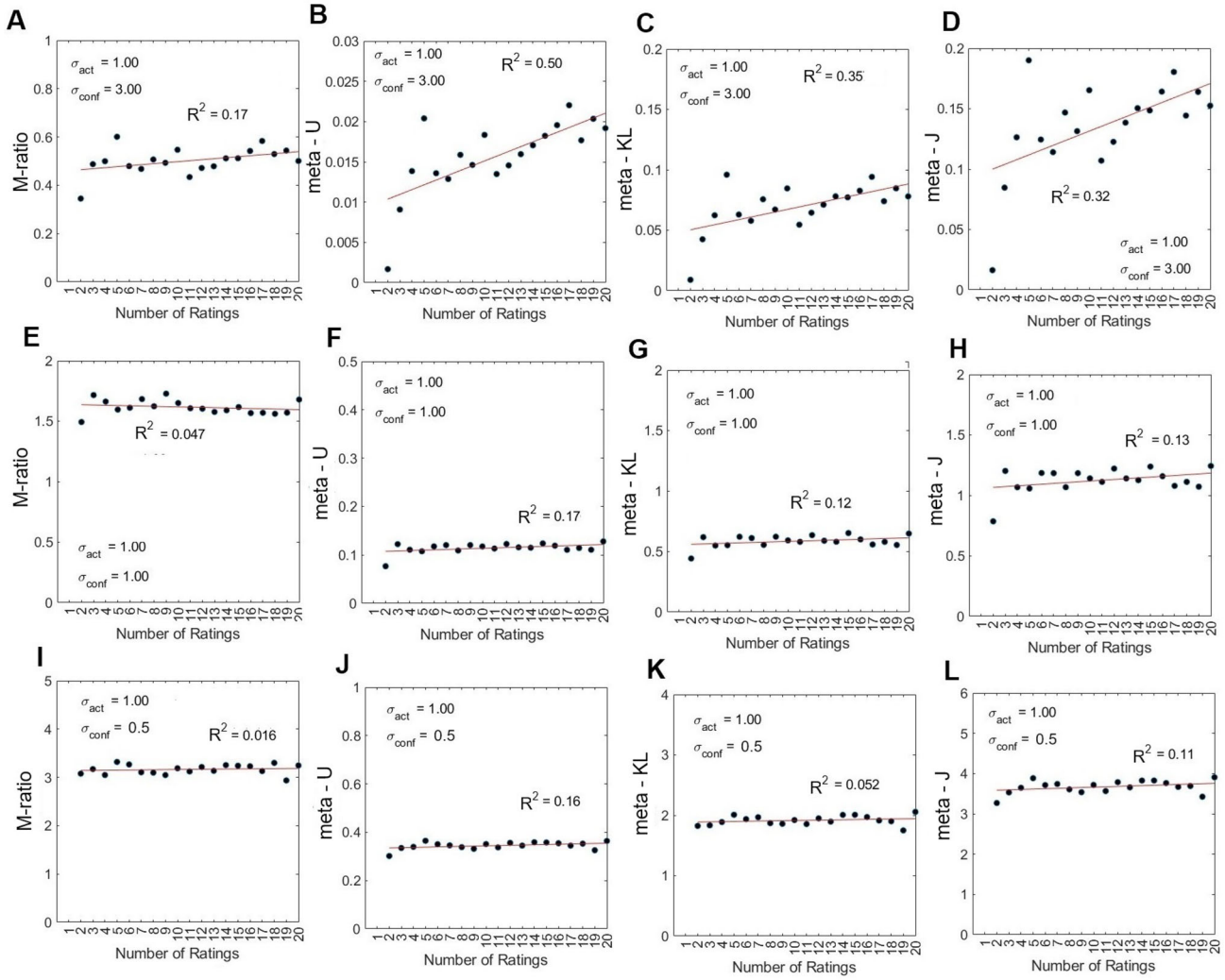


Fig. 15 On each row: $M-ratio, meta-\mathcal{U}, meta-\mathcal{KL}$ and $meta-\mathcal{J}$ as a function of number of confidence rating-buckets. **(A)**–**(D)** a second-order model with $\sigma_{conf} \gg \sigma_{act}$. **(E)**–**(H)** Second-order model with $\sigma_{conf} = \sigma_{act}$. **(I)**–**(L)** Second-order model with $\sigma_{conf} < \sigma_{act}$. The partition of posterior probability is as follows: 0–0.8, 0.8–0.875, 0.875–0.95, and 0.95–1. $\rho = 0.2$, $d' = 2$, 1000 trials were sampled in each simulation

gence-based measures – *meta-*$\mathcal{KL}$ and *meta-*$\mathcal{J}$ – and one mutual information measure, *meta-*$\mathcal{U}$. Table 1 provides a succinct summary of the information-theoretic measures, their formulas, and verbal descriptions of their corresponding quantities. These measures and their construct validity received substantial theoretical support via simulations. Although these measures capture different quantities, they are strongly correlated with one another, as well as with canonical signal detection theory (SDT) metrics such as *Meta-*d' and the M-ratio.

The information-theoretic measures possess several desirable properties: they are relative rather than absolute, and thus can adjust for type 1 performance (Maniscalco & Lau, 2014); they are computationally straightforward (Feldman, 2021); they do not rely on parametric assumptions; they are applicable to multi-alternative designs (Li & Ma, 2020); they are compatible with existing SDT frameworks (Fleming, 2017; Dayan, 2023); and they are invariant to the specific labels assigned to stimuli, responses, or confidence ratings (Shannon, 1948). Beyond these practical advantages, these measures offer novel insights into the mechanisms underlying metacognitive efficiency, due to the diverse ways in which they can be derived and interpreted.

Parametric versus non-parametric measures of metacognition

Several researchers (Mamassian & de Gardelle, 2022; Rausch et al., 2023; Rahnev, 2021) have strongly argued that metacognition should be framed exclusively in terms of pro-

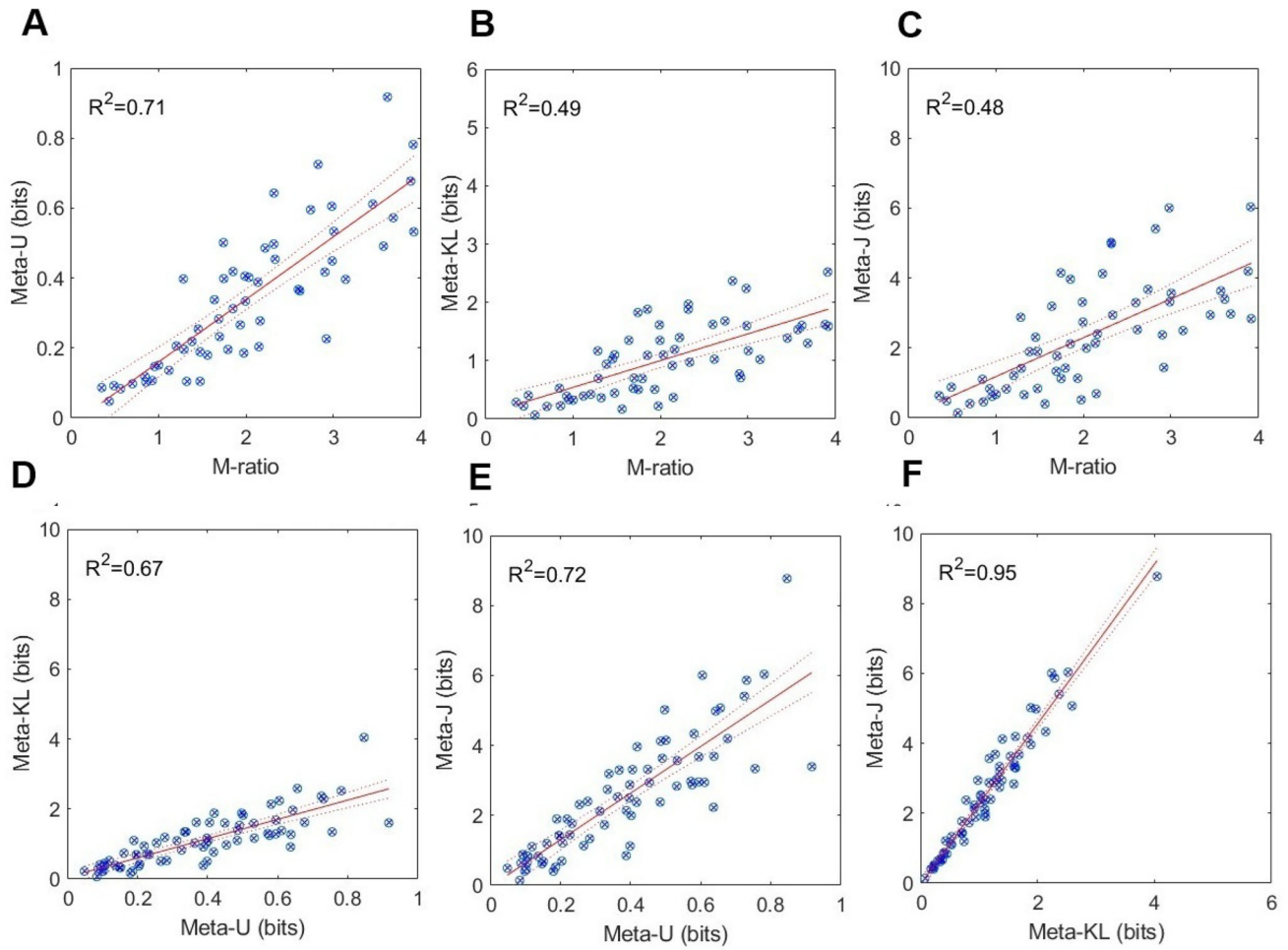


Fig. 16 Empirical results supporting the theoretical predictions developed in this paper. Figure reproduced from Fitousi (2025b), published under the terms of the Creative Commons Attribution License (CC BY 4.0). © 2025 by the authors; published by MDPI, Basel, Switzerland. Results from Experiment 2 of Fitousi (2025b), using the Kent Face Matching Task (KFMT): Associations between signal detection theory (SDT) measures and information-theoretic indices of metacognitive efficiency. Panels **(A)**–**(C)** show the relationship between M-ratio and the information measures meta-$\mathcal{U}$, meta-$\mathcal{KL}$, and meta-$\mathcal{J}$, respectively. Panels **(D)**–**(F)** show the relationship among the information-theoretic measures meta-$\mathcal{U}$, meta-$\mathcal{KL}$, and meta-$\mathcal{J}$. *Blue circles* represent individual data points; the *solid red line* indicates the best-fitting regression line; and *dashed red lines* represent the 95% confidence interval

cess models. According to this view, measures of metacognitive sensitivity and efficiency derived from parametric models are preferable to those based on non-parametric approaches. The rationale is that having a clear formulation of the underlying processes and constructs is critical for accurate estimation. While this approach is valid, it is worth noting that the information-theoretic framework advanced here provides a valuable and theoretically powerful complement. The proposed measures are parameter-free, yet they can be readily integrated with parametric and hierarchical frameworks such as those used in Bayesian modeling and hierarchical meta-d' estimation (Fleming, 2017; Guggenmos, 2021). Rather than being committed to any specific cognitive architecture, information-theoretic metrics maintain a level of abstraction that enables comparison across tasks, models, and populations. They also capture the core computational goal of metacognition – uncertainty reduction. The benefits of information-theoretic measures can be utilized in the emerging research on AI. A case in point is the need to estimate the metacognitive capacities of large language models (LLMs), and to compare their performance to that of humans (Steyvers & Peters, 2025). The psychological processes underlying LLMs are less well understood than those characterizing human cognition. In such cases, the application of abstract information-theoretic measures may be preferable to the use of principled parametric models originally designed to capture human cognitive processes.

The information-theoretic approach redefines metacognition not merely as introspection, but as a structured information transmission process. Every confidence report

Table 1 Summary of information-theoretic measures of metacognitive efficiency

Name	Formulas	Description
meta-$\mathcal{I}$ (Dayan, 2023)	$H_2(r\|a=-1) - H_2(r\|c,a=-1)$	Reduction in uncertainty about accuracy r given confidence c, using binary entropy.
meta-$\mathcal{I}'$ (adapted)	$I(r;c,s) = H(r) - H(r\|s,c)$	Mutual information between response r and joint stimulus-confidence (s,c); applicable to multi-alternative tasks.
meta-$\mathcal{U}$	$U(s{:}r{:}c) - U(s{:}r) = H(c) + H(s,r) - H(s,r,c) = I(c;s,r)$	Reduction in total correlation among s,r,c; reflects shared structure between stimulus, response, and confidence = mutual information between confidence c and joint stimulus-response (s,r).
meta-$\mathcal{KL}$	$\mathcal{KL} = \sum_y P(y\|\text{Correct}) \log_2 \frac{P(y\|\text{Correct})}{Q(y\|\text{Incorrect})}$	Asymmetric KL divergence between confidence distributions conditioned on correct vs. incorrect responses.
meta-$\mathcal{J}$	$\mathcal{J} = \sum_y \left[P(y\|\text{Correct}) \log_2 \frac{P(y\|\text{Correct})}{Q(y\|\text{Incorrect})} + Q(y\|\text{Incorrect}) \log_2 \frac{Q(y\|\text{Incorrect})}{P(y\|\text{Correct})} \right]$	Symmetrized KL (Jeffreys divergence) between confidence distributions for correct vs. incorrect responses.

constitutes the output of a communication system involving three interacting agents: the stimulus source (e.g., the experimenter), the actor (the perceiver/decider), and the rater (the introspective evaluator, who may be internal or external to the actor). Each agent reduces uncertainty through its own channel: the actor forms an internal estimate of the stimulus; the rater evaluates the internal state of the actor; and the experimenter manipulates task entropy and payoff structures (Killeen & Taylor, 2001; Garner, 1962). By treating the system in this way, we gain access to a myriad of theoretical results from information theory, such as channel capacity limits, rate–distortion trade-offs, and bottleneck phenomena (Berger, 2003; Fitousi, 2025a; Shannon, 1948; Sims, 2016; Tishby, Pereira, & Bialek, 2000). These concepts provide a powerful new lens for investigating the constraints that shape metacognitive behavior and its neural implementation.

Another key advantage of information-theoretic measures of metacognitive efficiency is that they are inherently constructed to avoid residual dependence on type 1 performance (Guggenmos, 2021). For example, Rausch et al. (2023) computed the M-ratio across several generative models and found that only in the case of the *independent truncated Gaussian model* was the M-ratio free from residual dependence on type 1 performance. By contrast, for the information-theoretic measures developed here, this independence is built in by design. Specifically, the computation of type 2 performance is not based on type 1 accuracy. Consider, for instance, *meta-*$\mathcal{U} = U(s{:}r{:}c) - U(s{:}r)$. This measure subtracts the pairwise dependence $U(s{:}r)$, which captures type 1 performance, from the total correlation $U(s{:}r{:}c)$, yielding a clean estimate of the mutual information between confidence and the joint stimulus–response variable. This property of independence from type 1 performance is illustrated in Fig. 9C.

A critical role for confidence granularity

A key insight from the information-theoretic approach to cognition (Algom & Fitousi, 2016; Garner, 1962, 1974; Melara & Algom, 2003; Melara & Mounts, 1993) is that the number of alternatives–whether in confidence ratings, stimuli, or responses–is a central determinant of behavior. For instance, a well-utilized eight-point confidence scale can convey more than twice the information of a four-point scale, assuming meaningful use of the full range. The same logic extends to the number of stimulus and response alternatives (Killeen & Taylor, 2001). Greater resolution enables observers to express subtler distinctions in certainty, increasing mutual information between confidence and task variables (in type 2 tasks) or between stimuli and responses (in type 1 tasks).

This perspective challenges a core assumption in standard process models of metacognition (e.g., SDT-based models), where confidence arises from a fixed-noise internal signal, and criteria placement is treated as arbitrary unless externally optimized (see Maniscalco et al., 2024). In such models, the number of confidence categories does not affect metacognitive sensitivity, which is tied to internal signal characteristics. However, this overlooks the communicative function of confidence ratings. If finer-grained scales allow a more expressive mapping from internal states to observable

responses, then granularity directly affects information transmission. The better a rater can externalize uncertainty, the higher the observed metacognitive efficiency. Similarly, more stimulus or response alternatives enable richer encoding of perceptual information, boosting type 1 discriminability (Hake & Garner, 1951).

From an information-theoretic perspective, assuming that confidence and decision are based on the same evidence, the idea of a hyper-sensitive rater (Maniscalco & Lau, 2014) (e.g., M-ratio > 1) is problematic, as it would imply a violation of the data processing inequality. However, I acknowledge that in some experimental designs (Maniscalco & Lau, 2014; Pleskac & Busemeyer, 2010), additional sensory or mnemonic information may become available between the decision and the confidence judgment. In such cases, what appears as hyper-sensitivity may reflect access to a richer evidence stream, rather than a paradox or mismeasurement. I view this as an important qualification, and encourage future work to use information-theoretic tools to explicitly model such multi-stage evidence accumulation.

Importantly, the present information-theoretic framework aligns with the taxonomy proposed by Shekhar and Rahnev (2021b), which identifies multiple sources of metacognitive inefficiency – ranging from degraded sensory input to noisy transformation functions and response variability. The information-theoretic approach partitions total uncertainty into *structural* constraints imposed by the task and *individual* constraints related to encoding, retention, or retrieval.

Structural constraints

These constraints are often inadvertently built into the experimental structure, and they are imposed by the number of alternatives assigned to each variable. For example, a 2-alternative forced choice (2AFC) task inherently limits stimulus entropy to 1 bit. Even if a four-point confidence scale is added, which could in theory convey 2 bits of information, the maximum joint entropy remains limited by the structure of the task. Unless task variables (e.g., stimulus alternatives, response options) are made more complex, the confidence channel cannot exceed this informational ceiling. Specifically, the amount of metacognitive information one can extract – formalized in the present paper as $I(c; s, r)$, the mutual information between confidence c, stimulus s, and response r – is limited from above by $\min[H(c), H(s, r)]$. The practical implication is thus to better assess the limits of metacognitive performance, experimenters must increase granularity along all task dimensions: stimulus, response, and rating scales. Otherwise, structural bottlenecks will mask the true expressive potential of metacognitive systems.

Individual constraints

In addition to structural limitations, individuals exhibit cognitive capacity limits due to memory, attention or other psychological constructs that can constrain metacognitive performance. Classic research on perceptual categorization shows that as the number of alternatives increases, information gain in type 1 decisions initially rises but then levels off due to processing limits (Miller, 1956; Garner & Hake, 1951). For instance, Hake and Garner (1951) demonstrated that even with 15 stimulus alternatives, observers do not exceed 3.3 bits of transmitted information. This suggests that observers can distinguish between roughly 10–11 options. In the present framework, this means that regardless of task information, the amount of information that an individual can extract and express is limited by personal capacity. Formally, this yields a saturation function (see Fig. 17): $I(S; R) = \min[H(S), K]$, where $H(S)$ is the entropy of the stimulus set and K is the observer's cognitive capacity. Beyond this point, increasing stimulus complexity will not provide any additional gains in mutual information, because the bottleneck is internal rather than structural. This has direct consequences for experimental design and interpretation. Observed metacognitive inefficiency might stem from a mismatch between task complexity and observer capacity. Some observers may benefit from simpler tasks with lower entropy, allowing their full capacity to be expressed, while others may underperform in high-complexity tasks due to saturation.

The individual constraints have important implications for group-level decision-making. In collective contexts such

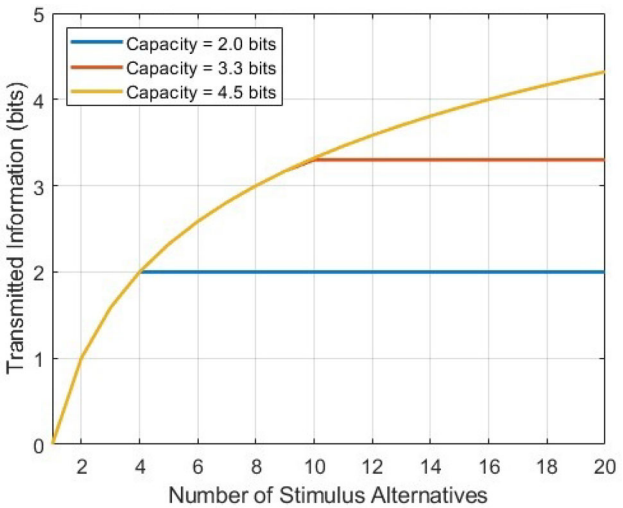


Fig. 17 Mutual information as a function of stimulus alternatives for actors with different levels of capacities. Information transmission increases with stimulus uncertainty until reaching individual capacity limits, after which it levels off. This illustrates cognitive saturation even under optimal task conditions

as group reasoning or social prediction (Bang et al., 2017), differences in individual capacity may lead to unequal contributions to group confidence calibration.

Recommendation

- Design metacognitive tasks with multiple stimuli and response alternatives to enable richer representations and more accurate estimation of information flow.
- Use confidence scales with as many levels as participants can reasonably handle, but ensure the number of confidence levels does not exceed the number of stimulus or response alternatives to avoid structural mismatches.
- Consider the possibility that observed metacognitive inefficiencies may stem from structural constraints of the stimulus-response space or individual cognitive limitations (e.g., memory, attention, internal noise), not just genuine deficits.
- Confidence ratings act as messages from rater to experimenter; maximizing informativeness requires participants to use a broad range of confidence categories.
- Narrow or low-entropy use of confidence scales reduces the capacity of this communication channel, representing a structural bottleneck in metacognitive information transmission.
- Extreme values of metacognitive efficiency (e.g., hypersensitivity) can reflect lawful outcomes of actor-rater communication dynamics.

Appendix A: Equivalence of $U(c:s,r)$ and $I(c;s,r)$

The measure $U(c:s,r)$, defined within Garner's multivariate dependence framework, was introduced in the main text as the difference:

$$U(c:s,r) = U(s:r:c) - U(s:r), \tag{28}$$

where $U(s:r:c)$ quantifies the total dependence among the triplet (s,r,c), and $U(s:r)$ captures the pairwise dependence between stimulus and response. Although $U(c:s,r)$ is not originally framed as mutual information, I show here that it is mathematically equivalent to the classical mutual information $I(c;s,r)$. Garner (1962) defined multivariate dependence as the difference between the maximum entropy under independence and the observed joint entropy. For three variables:

$$U(s:r:c) = H(s) + H(r) + H(c) - H(s,r,c), \tag{29}$$

and for two variables:

$$U(s:r) = H(s) + H(r) - H(s,r). \tag{30}$$

Plugging Eqs. 29 and 30 into Eq. 28, we find:

$$\begin{aligned} U(c:s,r) &= [H(s) + H(r) + H(c) - H(s,r,c)] - [H(s) \\ &\quad + H(r) - H(s,r)] \\ &= H(c) + H(s,r) - H(s,r,c). \end{aligned} \tag{31}$$

The mutual information between confidence and the joint stimulus-response pair is:

$$I(c;s,r) = H(c) + H(s,r) - H(s,r,c). \tag{32}$$

This is precisely the same quantity as in Eq. 31. Therefore:

$$U(c:s,r) = I(c;s,r). \tag{33}$$

Author Contributions D.F. has developed the ideas, performed the simulations and wrote the paper.

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Availability of data and materials All data, codes, and supporting materials can be downloaded from https://osf.io/z5swc/

Code availability (software application or custom code) These are available here https://osf.io/z5swc/

Declarations

Conflicts of Interest I have no conflicts of interest. I have no competing interests.

Ethical Approval This is not relevant to the present study.

Consent to Participate This is not relevant for the present study.

Consent for Publication This is not relevant for the present study.

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